THE SPECTRAL SHORTCUT: HOW A FAST CONTEXTUAL PATHWAY SUPPRESSES SPECTRAL LEARNING IN SERIAL SPECTRAL–SPATIAL MODELS

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ABSTRACT

We study how relative module training rates affect spectral adaptation in serial spectral–spatial models, in which a small encoder compresses each pixel's spectrum and a wide spatial network reads the encoded neighbourhood. An exactly solvable serial logistic model bounds the spectral encoder's update at matched fit by a fraction $1/(1 + Mv_0^2)$ of the update a spectral-only model needs, establishes contextual-reversal failure under explicit finite-width conditions, and shows through a $M^{-1/2}$-normalized readout that replicated-readout width acts through the training metric; normalization removes the width dependence but need not prevent the failure. In a nonlinear synthetic model with two cues of matched signal-to-noise ratio, slowing the spatial head by $256\times$ raises the linear accessibility of the spectral cue at matched training loss from 0.66 to 0.85 while the trained classifier remains strongly context-reliant (reversal accuracy 0.12 to 0.14); retraining the head on decorrelated context from the frozen encoders then recovers shifted accuracy of 0.75–0.84 from the slow-head encoders against 0.54–0.62 from the fast-head ones, no better than random encoders. Experiments using measured tissue spectra with constructed context distinguish high and low initial spectral accessibility, show strong contextual reliance in both, and show a weaker rate response at high contextual amplitude. The results separate adaptation of the encoder, accessibility of the spectral cue, reliance of the fitted classifier, and what a specified retraining recovers.

1 INTRODUCTION

Infrared and quantum-cascade-laser histology classify tissue pixel by pixel from a spectrum of several hundred channels, and the models that do so are serial: a small encoder compresses each pixel's spectrum, and a much larger spatial network reads the encoded neighbourhood (Berisha et al., 2019; O'Leary et al., 2026). The same shape appears wherever a per-token encoder feeds a contextual model. Practitioners often pretrain or freeze the spectral stage, and a companion tokenization study (anonymous, under review) reports a frozen random encoder competing with a jointly trained one. Such comparisons do not settle whether joint training starves the encoder: the spatial stage can predict the label from the neighbourhood, and a model that learns to do so may never need the spectrum.

This paper asks how the relative training speed of the two modules affects what the encoder learns, and keeps four measurements apart that are easily conflated: adaptation of the encoder, linear accessibility of the spectral cue in its output, reliance of the trained classifier on context, and what a specified retraining procedure can recover afterwards. We prove one instance exactly: in a linear-encoder model with a replicated contextual readout, a faster contextual pathway suppresses the encoder's update at matched fit, the fitted model fails under contextual reversal under explicit finite-width conditions, and readout normalization removes the width dependence (Section 3). We then measure, with interventions on training speed, how each of the four quantities responds in a nonlinear convolutional model on synthetic data (Section 4) and with measured tissue spectra in constructed neighbourhoods (Section 5), in each case against a control that removes the proposed cause.

Contributions.

- **Theorem 1.** For the serial cross-entropy model with M replicated contextual readouts, gradient flow from general initialization moves the spectral coefficient at the matched fitting margin by at most a fraction $1/(1 + Mv_0^2)$ of the update a spectral-only model needs; the fitted model misclassifies every example under contextual reversal when the initial spectral coefficient is below half the margin and $Mv_0^2 > m/(m-2a_0)$, with expected error tending to $\Phi(m/2\sigma)$ over isotropic Gaussian initialization as $M \to \infty$; and a $M^{-1/2}$-normalized readout removes the width dependence, though it need not remove the failure.
- **Synthetic intervention.** In a ReLU convolutional head over a linear encoder, with two cues of matched signal-to-noise ratio but unequal initial accessibility, an informative context suppresses the encoder's spectral learning at matched fit at every width. Slowing the whole head by $256\times$ raises the spectral cue's linear accessibility from 0.66 to 0.85, monotonically in every seed, while the trained classifier's reversal accuracy moves from 0.12 to 0.14; a normalized readout reduces the suppression at large width; an uninformative context removes it. Retraining a fresh head on decorrelated context from the frozen encoders recovers shifted accuracy of 0.75–0.84 after the slow head and 0.54–0.62 after the fast one, no more than from random encoders.
- **Real spectra.** With measured centre spectra and their recorded labels in constructed neighbourhoods, informative-context arms are strongly context-reliant in both a high- and a low-accessibility regime; in the low-accessibility regime the encoder's adaptation is limited relative to the uninformative-context control, with a small rate response at the larger cue amplitude and a larger one at the smaller amplitude, and more than 99.9% of the net encoder update lies in the constructed-cue span. A readiness scan of real class pairs shows that preprocessing and bottleneck decide which regime a pipeline is in.
- **Scope.** We report why scalar curvature on a production model does not identify the mechanism and why an earlier apparent freezing advantage was protocol-confounded (Section 6, Appendix E); no mitigation is recommended.

Related work. That one predictive feature can suppress the learning of another is gradient starvation (Pezeshki et al., 2021), and a preference for simpler features that fails under shift is simplicity bias (Shah et al., 2020); modality competition in multimodal training (Wang et al., 2020; Peng et al., 2022; Huang et al., 2022; Du et al., 2023) studies the same imbalance between input streams. That core features can be present in a representation while a spurious feature is used, and that last-layer retraining can then restore robustness, is the observation of Kirichenko et al. (2023); our addition is that the relative training rate of the modules at the original fit changes what such retraining can recover, measured at matched fit against a control. The mathematics uses the balance invariant of gradient flow in layered linear models (Du et al., 2018; Saxe et al., 2014; Yun et al., 2021) and sits next to lazy-training analyses in which a wide component barely moves (Chizat et al., 2019; Yang & Hu, 2021). Curvature and its growth with width and normalization (Karakida et al., 2019; van Laarhoven, 2017) are the reason we measure learning outcomes rather than curvature (Appendix E); frozen or retrained feature extractors in practice (Zhang et al., 2022; Coil & Cheney, 2025) are the setting, not a remedy we evaluate.

2 Model, task, and what counts as failure

Serial spectral–spatial model. An input is an image whose pixel p carries a spectrum $x_p \in \mathbb{R}^S$ and a label $y_p \in \{-1, +1\}$. A spectral encoder $f_{\boldsymbol{\theta}} : \mathbb{R}^S \to \mathbb{R}^K$ acts on each pixel separately, and a spatial model $g_{\boldsymbol{\phi}}$ of width M reads the encoded neighbourhood of p to produce a logit F_p. The encoder is linear, $f_{\boldsymbol{\theta}}(x) = Wx$ with $W \in \mathbb{R}^{K\times S}$, in the theorem and in the experiments; the spatial model is a replicated readout in the theorem and a ReLU convolutional head in the experiments. Training minimizes the mean margin loss $\mathcal{L} = \frac{1}{N}\sum_p \log(1 + e^{-y_p F_p})$ by gradient flow or plain gradient descent with a base rate η; the baseline uses that one rate for every parameter, and the interventions multiply the rate of a named block by a stated factor, so that every difference in training speed between the modules is either a property of the model or an explicit, declared multiplier.

Two cues. The training distribution carries the label in two places. The *spectral cue* lies in the pixel's own spectrum along a direction the encoder must discover: a randomly initialized encoder

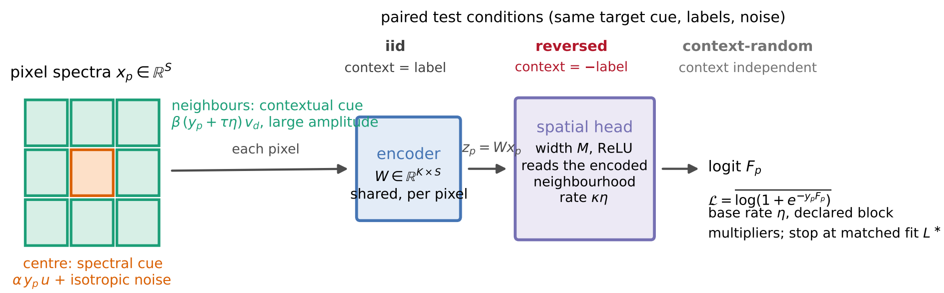

Figure 1: The serial model and the two cues. Each pixel's spectrum passes through the shared encoder; the spatial model reads the encoded neighbourhood (a 5×5 receptive field from two 3×3 convolutions in Experiment 2; one nine-spectrum patch in Experiment 3). The contextual cue is placed in the eight nearest neighbours along large-amplitude directions that a random encoder already passes; the spectral cue sits in the pixel's own spectrum along a direction of small amplitude and, in the low-accessibility settings, requires encoder alignment. Evaluation under iid, reversed and context-random contexts measures reliance and failure.

attenuates it. The *contextual cue* lies in the neighbourhood along directions of large amplitude: a randomly initialized encoder passes it, so a trained readout can use it at once. The two cues are calibrated to matched discriminant accuracy on calibration data (a matched signal-to-noise construction in the synthetic case; a training-side estimate on real spectra), so that a preference for one cue is not simply a preference for the more predictive one. The theorem allows any (a_0, v_0); the experiments select its regime of interest, a small initial spectral contribution and an initially accessible contextual one, and measure that accessibility asymmetry directly (Section 4).

Matched fit, spectral learning, reliance, failure. We compare models at their first attainment of a prespecified training-fit threshold: a common margin in the solvable model and mean training loss in the experiments. Spectral learning means adaptation of the designated spectral coefficient in the theorem and increased independently measured linear accessibility of the centre's spectral cue in the experiments. We measure contextual reliance by changing the assigned contextual signal while preserving the target's spectral cue, label and noise, and recording the resulting change in predictions or risk. We call that reliance a failure when the shifted risk exceeds that of a comparator of the same architecture trained with an uninformative (decorrelated) context to the same fitting threshold. The theorem supplies an exact reversal failure against a spectral-only model; the noisy experiments measure the corresponding risk difference against that trained comparator, without assuming a zero-error reference. Four measurements are kept distinct throughout: adaptation of the encoder, linear accessibility of the spectral cue on clean spectral inputs (the probe), use of context by the original predictor, and performance after an explicit retraining procedure.

Test conditions. Every evaluation family is generated from the same labels and noise in several ways: *iid* (context agrees with the label as in training), *reversed* (the assigned context carries the opposite label) and *context-random* (the assigned context carries an independent label, so context is present but uninformative). In the synthetic experiment the intervention changes the contextual feature field at every pixel while the target's spectral component, label and noise are shared; in the real-spectrum experiment each centre spectrum is held fixed and only its neighbours change. Accuracy under reversal measures reliance; accuracy under context-random measures what the original predictor does from the spectrum in the presence of context.

3 Main result: selective spectral learning at matched fit

We state the mechanism in the simplest serial model that has both cues, a shared encoder, and a replicated contextual readout. Training inputs are two sites: the pixel's own spectral coordinate S and a contextual coordinate C, both equal to the label. The encoder $W = (a, v)$ reads each site with one coefficient; the head adds the spectral coordinate directly and the contextual one through M

replicated readouts. Replication is the only way width enters. The statement combines the serial instance, its general-initialization corollary and the readout-normalization proposition of Appendix A.

Theorem 1 (Selective spectral learning at matched fit)**.** *Let Y be uniform on $\{-1, 1\}$, $x_1 = (S, 0)$, $x_2 = (0, C)$, and $S = C = Y$ in training. The shared encoder $W = (a, v)$ produces $z_1 = aS, z_2 = vC$; the head is $F = z_1 + bz_2$, $b = \sum_{j=1}^{M} \beta_j$. For integer $M \geq 1$, train (W, β) by unit Euclidean gradient flow on $\mathcal{L} = \mathbb{E} \log(1 + e^{-YF})$, without other optimizer terms, from fixed $a_0 \in \mathbb{R}$, $v_0 \neq 0$, $\sum_j \beta_{j0} = 0$. Fix $m > 0$ and $T_m = \inf\{t \geq 0 : q_J(t) \geq m\}$, $q_J = a + bv$; thus $T_m = 0$ if $a_0 \geq m$.*

(a) Suppression. *If $a_0 < m$, set $\Delta = m - a_0$, $u_M = a(T_m) - a_0$. Then $T_m < \infty$ and*

$$0 < \frac{u_M}{\Delta} \leq \frac{1}{1 + Mv_0^2}, \qquad Mu_M \longrightarrow \frac{\Delta}{v_0^2} \quad (M \to \infty).$$

The spectral-only head $F = aS$ requires update Δ. If q_F instead trains only β from the same initialization with W frozen, then $q_J \geq q_F$ on $[0, T_m]$ and $\sup_{0 \leq t \leq T_m} (q_J - q_F) \leq C_0/(p_0 M v_0^2)$, where $C_0 = 1 + 2\Delta^2/v_0^2$ and $p_0 = (1 + e^{-a_0})^{-1}$.

(b) Reversal. *On $S = Y, C = -Y$, the stopped margin is $2a(T_m) - m$ when $a_0 < m$, and a_0 otherwise. Error is one if $a_0 < m/2$ and $Mv_0^2 > m/(m - 2a_0)$; it is zero for $a_0 \geq m/2$. The spectral-only comparator has zero error for every a_0. For $(a_0, v_0) \sim \mathcal{N}(0, \sigma^2 I_2)$, $\sigma > 0$, with zero-sum readouts and the same stopping rule, $\lim_{M\to\infty} \mathbb{E}_{W_0} \mathrm{Err}_{\mathrm{rev}} = \Phi(m/(2\sigma))$, where Φ is the standard normal CDF.*

(c) Normalization. *Replace b by $M^{-1/2} \sum_j \gamma_j$, with $\sum_j \gamma_{j0} = 0$ and the same (a_0, v_0) and unit rates. The aggregate (a, v, b) dynamics equal those at $M = 1$ for every width.*

Remark 1 (Special initialization). For $(a_0, v_0) = (0, 1)$, the scalar-logit GGN blocks $G_{\boldsymbol{\theta\theta}}, G_{\boldsymbol{\phi\phi}}$, $\boldsymbol{\theta} = (a, v)$, $\boldsymbol{\phi} = \beta$, give $\mathcal{D}_{\mathrm{curv}}(0) = M$ (generally Mv_0^2), and $\Psi(m)/(M + 1 + 2m^2) \leq T_m \leq \Psi(m)/(M + 1)$, where $\Psi(m) = m + e^m - 1$.

Proof sketch. Dividing the flow by $\dot{a} = \varrho > 0$, $\varrho = (1 + e^{q_J})^{-1}$, gives $dv/da = b$ and $db/da = Mv$, hence with $u = a - a_0$

$$v = v_0 \cosh(\sqrt{M}u), \qquad b = \sqrt{M} v_0 \sinh(\sqrt{M}u), \qquad q_J = a_0 + u + \tfrac{\sqrt{M} v_0^2}{2} \sinh(2\sqrt{M}u),$$

the balance invariant of gradient flow (Du et al., 2018). The fitting equation $q_J(u_M) = m$ is strictly increasing in u and $\sinh x \geq x$ gives the update bound and its limit; a transformed-margin comparison with the frozen path gives the finite-horizon gap. The sign of the reversed margin $2a(T_m) - m$ and dominated convergence give (b). For (c), differentiating the aggregate readout gives $\dot{b} = v\varrho$, independent of M. Full proofs are in Appendix A.

What the theorem says, and what it does not. Replication acts as an effective learning rate on the contextual readout. At the matched margin the spectral coefficient has moved by at most a fraction $1/(1 + Mv_0^2)$ of the update the spectral-only model needs; the coefficient itself can be informative at initialization and remain so. It is the *adaptation* that is suppressed. When the initial spectral coefficient is below half the margin and the width is large enough, $Mv_0^2 > m/(m - 2a_0)$, the fitted joint model relies on context strongly enough that contextual reversal misclassifies every example, while a spectral-only comparator trained to its own hitting time of the same margin is correct; at small Mv_0^2 the fitted model can be correct even with $a_0 < m/2$, so the failure is a finite-width statement, not a universal one. The spectral-only comparator is trained to its own hitting time; the frozen comparator is evaluated at the joint path's elapsed times through T_m. Part (c) removes the width dependence only: the normalized system is the $M = 1$ system, which for $(a_0, v_0) = (0, 1)$ still fails under reversal for every $m > 0$. Finally, training each contextual readout at rate κ instead of unit rate gives $\dot{b} = \kappa M v \varrho$ and the bound $1/(1 + \kappa M v_0^2)$: the relative training speed of the contextual pathway, not width as such, is the competition parameter the experiments manipulate.

Numerical check. The closed forms agree with a full $(2 + M)$-parameter integration over 525 initialization–width–margin cases to a relative discrepancy of 1.3×10^{-11} in the endpoint quantities, with no bound violated, and the Gaussian sweep approaches its limit $\Phi(1)$ (0.8207, 0.8380, 0.8397 at $M = 64$, 1024, 16384; Appendix B). This checks the algebra, not the claim beyond the model.

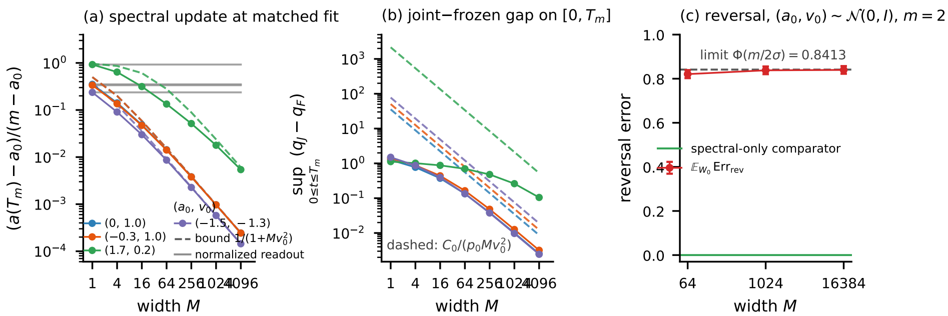

Figure 2: The exact mechanism (Theorem 1; $m = 2.9$ unless stated). (a) Spectral update fraction u_M/Δ at the matched margin against width M for four initializations (a_0, v_0) (solid) with the bound $1/(1 + Mv_0^2)$ (dashed) and the normalized readout of part (c) (grey, flat at the $M = 1$ value). (b) Joint-minus-frozen margin gap $\sup_{0\le t\le T_m}(q_J - q_F)$ with the bound $C_0/(p_0Mv_0^2)$ (dashed). (c) Expected reversal error over $(a_0, v_0) \sim \mathcal{N}(0, I_2)$ against M for $m = 2$, with the limit $\Phi(m/2)$; the spectral-only comparator has error zero.

4 Exact-model check and synthetic intervention

The exact model is verified numerically in Section 3; the question here is whether the mechanism survives when the cues are noisy, the routing is learned and the head is nonlinear.

Experiment 2: design. Pixels carry i.i.d. labels $y_p \in \{-1, +1\}$ on 16×16 images with $S = 256$ spectral channels, so the spectral cue cannot be denoised by averaging neighbours. The spectral cue is $\alpha y_p u$ in the pixel's own spectrum, buried in isotropic noise of unit variance, with u a fixed unit direction the encoder must discover. The contextual cue places y_p in each of the eight neighbours of p, along eight fixed orthogonal directions $v_1, \dots, v_8 \perp u$ of amplitude $\beta = 5$ with context noise $\tau\eta$: the information about y_p sits only in the neighbourhood, because p's own v-coordinates carry its neighbours' labels. Matched signal-to-noise: $\alpha = 1.645$ gives an oracle spectral accuracy of 0.95, and $\tau = 1.70$ is calibrated so that the oracle contextual accuracy (a linear discriminant on the 3×3 window of the eight projections) is 0.950; both values are recorded before any training run (Appendix C). Readiness: through a random encoder $W \in \mathbb{R}^{12\times 256}$ the same discriminant reads the contextual cue at 0.905–0.914 and the spectral cue at 0.608–0.653 (three seeds), because the random projection preserves the signal-to-noise ratio of a cue whose noise shares its direction and attenuates by about $\sqrt{K/S}$ one whose noise is isotropic. This is the nonlinear analogue of (a_0 small, $v_0 = O(1)$): the contextual cue is initially accessible to a trained readout; the spectral cue is substantially less accessible. The model is the linear encoder followed by a ReLU convolutional head (3×3 convolution to width M, ReLU, 3×3 convolution to one logit, circular padding; a 5×5 receptive field), trained by full-batch gradient descent on 256 images (65,536 pixels) with base rate $\eta = 10^{-3}$, chosen as the largest rate with a monotone loss at the widest head; the baseline applies it to every parameter, the rate interventions multiply a named block's rate. Runs are compared at the first step with training loss below $L^* = 0.30$, with further snapshots at $0.6, 0.5, 0.4, 0.2, 0.15, 0.10$.

Arms and measures. Standard parameterization at widths $M \in \{2, 4, 8, 32, 128, 512, 2048\}$; a normalized readout ($M^{-1/2}\gamma$ with $\gamma = O(1)$ and the same function at initialization, the control of Theorem 1(c), which under plain gradient descent equals a readout trained at rate $1/M$); training with an uninformative context (the same cue carrying an independent label: the proposed cause removed) at $M \in \{2, 8, 128, 512\}$; a readout learning-rate multiplier and, following the design review, a whole-head multiplier $\kappa \in \{1, 1/16, 1/256\}$ at $M = 32$, which in the theorem gives the bound $1/(1 + \kappa M v_0^2)$; and a frozen encoder. Three seeds each. At every snapshot we record the probe (a discriminant on the encoder output of clean spectral-only inputs, fitted and evaluated on disjoint independent sets; initial values 0.647, 0.658, 0.615, reported as the gain over the run's own initial value), its exact population counterpart $h_u(W) = \|P_{\mathrm{row}(W)}u\|^2$ (optimal probe accuracy $\Phi(\alpha\sqrt{h_u})$; 0.055, 0.060, 0.029 at initialization), the encoder's displacement, and accuracies on paired test sets sharing labels and noise (iid, reversed and context-random change only the contextual field; spectral-only removes it; context-only removes the spectral signal). A recovery procedure was fixed before any shifted result was inspected: the encoder is frozen at its L^* checkpoint, a fresh head with a paired initialization is trained on newly sampled context-random images for $20{,}000$ steps, and the final head is evaluated on the same paired test family.

Results: suppression. In the standard-parameterization baseline every model with an informative context leaves the encoder's spectral accessibility near its initial value at L^*: the probe gain is 0.04 at $M = 2$ and 0.002 at $M = 2048$ (means over three seeds; h_u $0.09 \to 0.05$ against a mean 0.048 at initialization), accuracy under reversal is 0.12–0.13, and accuracy with an uninformative context is at chance (0.50). The same model trained with an uninformative context aligns its encoder before reaching the same loss (probe gain 0.25–0.30; h_u 0.56–0.86) and survives reversal (0.88–0.90). A frozen encoder gives reversal accuracy 0.12–0.14 (Table 4).

Results: the registered predictions. The original width-based prediction was not met at $L^* = 0.30$: on the original five widths only one seed shows the required rank correlations for both alignment and reversal accuracy. On those widths, the normalized readout's alignment and probe flatness met the stated two-of-three-seed criterion, but its reversal flatness did not; its high-width alignment improvement over the standard readout held in every seed. On the amended control grid, the uninformative-context arm failed both the energy-fraction cutoff and the flatness requirement, although it produced substantially higher probe accuracy than informative-context training at every width. The readout-rate sweep supported the alignment prediction and changed reversal accuracy in the opposite direction; frozen encoders remained far below chance under reversal; the whole-head-rate prediction added at the design review held for probe accessibility and row-space retention in every seed, while its reversal rank criterion failed in one seed. Low primary-threshold gains with the stronger labelled secondary trend at loss 0.15 are consistent with contextual fitting preceding spectral adaptation, and neither falsify nor confirm competition (criterion table in Appendix C).

Results: where the width and speed dependence appears. At the labelled secondary threshold 0.15, spectral learning decreases with width in every seed (probe gain 0.22 at $M = 2$, 0.12 at $M = 8$, 0.06 at $M = 128$, 0.03 at $M = 2048$; Spearman -0.86, -0.96, -0.86), and the normalized readout is less suppressed at large width (0.10 at $M = 2048$ against 0.03; Table 5). The whole-head multiplier moves the same quantity at L^* itself: reducing the head's rate by 256 raises mean probe accuracy from 0.662 to 0.851 (paired gains 0.195, 0.178, 0.192) and h_u from 0.067 to 0.406, monotone in every seed (Table 7), in the direction of the theorem's bound $1/(1 + \kappa M v_0^2)$, while remaining below the uninformative-context control (0.89–0.94). Two further runs at $\kappa \le 1/4096$ on one seed did not reach L^* within the step budget (final losses 0.32 and 0.35, $h_u = 0.69$–0.70); they are reported as budget endpoints, not matched-fit points. The readout-only multiplier acts in the same direction but weakly (Table 8).

Results: the original predictor stays context-reliant. Across every matched informative-context run, accuracy under reversal at L^* lies between 0.10 and 0.15 and accuracy with an uninformative context at chance, for every width, the normalized readout, the readout multiplier and the whole-head multiplier; the rate change that raises the probe by 0.19 raises reversal accuracy by 0.02 ($0.121 \to 0.143$). Among the tested interventions, only training without an informative context removes the reliance. The trained classifier's use of context and the encoder's accessibility thus respond very differently to the same intervention. We do not read a toy margin decomposition into the CNN; the measured outcomes suffice.

Results: what retraining recovers. The recovery procedure makes the accessibility difference operational. From the $\kappa = 1$ encoders the retrained head reaches accuracy 0.62, 0.59, 0.54 under reversal (0.61, 0.59, 0.55 with context-random, 0.59 iid), no better than the same procedure on the random initial encoders (0.60, 0.59, 0.53); from the $\kappa = 1/256$ encoders it reaches 0.84, 0.77, 0.75 (0.84, 0.77, 0.76; 0.79 iid), while the original classifiers of the two arms had the same poor shifted performance (0.12–0.15; Table 9). Within this protocol the contextual pathway's relative rate at the original fit thus changed what retraining could recover by about 0.2; no retraining reached the 0.10 loss target, so this concerns the specified procedure and budget, not every possible repair.

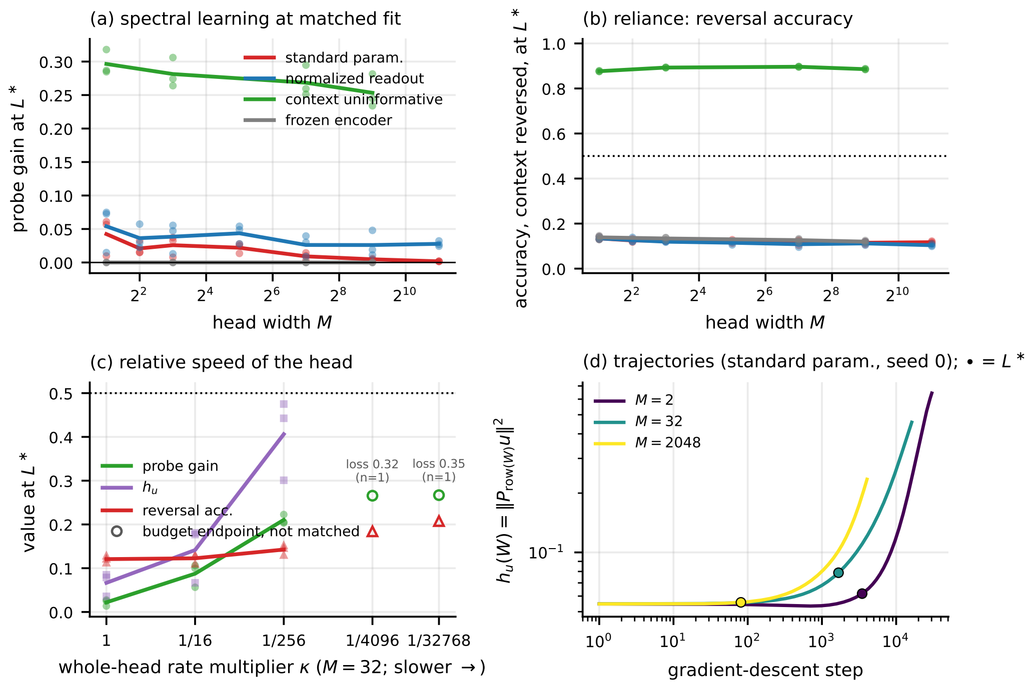

Figure 3: Experiment 2 at matched fit $L^* = 0.30$ (points are seeds, lines means). (a) Spectral probe gain against head width by arm. (b) Accuracy under context reversal. (c) Whole-head rate multiplier κ at $M = 32$: probe gain, h_u and reversal accuracy; open markers are single-seed budget endpoints of runs that did not reach L^* (losses shown), not matched-fit points. (d) h_u along training for three widths, L^* crossing marked.

Amendments. Widths 2 and 4, the fractional readout multipliers and the whole-head arm were added after a one-seed pilot and the design review, before their runs; the κ extension below $1/256$ was motivated by the observed reversal outcome; the recovery procedure was specified after the first results and before its runs (Appendix C).

5 Real centre spectra with constructed context

What this experiment is. We test contextual competition using measured centre spectra and their recorded labels, with a synthetic class-associated spatial intensity pattern along training-estimated principal directions in the neighbourhood. It is a real-centre-spectrum experiment with constructed context, not a validation on natural tissue context: it asks whether the mechanism operates when the spectral cue is a real, high-dimensional class contrast, and it cannot show that natural neighbourhoods produce it.

Data and readiness. Breast-tissue QCL micro-array cores, fold 0 of a five-fold patient-level split; the binary task Cancer Epithelium versus Cancer-Associated Stroma on 942-channel pixel vectors; $24{,}000$ class-balanced training centres; evaluation on held-out validation patients, with test patients reported separately (Appendix D). A readiness scan (Table 11) shows that with per-feature standardization a random encoder $W \in \mathbb{R}^{12\times 942}$ already reads this contrast at 0.955 against a calibration discriminant of 0.969, because the class-mean contrast lies along the top principal directions; PCA whitening estimated on the training patients reduces the random-encoder probe to 0.54 while the discriminant stays at 0.944. We therefore run two regimes on the same pixels and labels, *high* and *low initial probe accessibility* (standardized and whitened inputs). They are an empirical analogue of the theorem's adaptation question, not instances of the theorem: a probe fitted on the encoder output does not measure the spectral contribution to a random head's margin, so neither regime is identified with a case of Theorem 1.

Construction. Each patch is a real centre with eight real donor spectra drawn independently of the centre label from the same split side; donor d carries $\gamma_d(c\,s + \tau\eta_d)$ along the dth principal coordinate, where c is the centre's label, $s = \pm 1$ or c is replaced by an independent label, and γ_d is large relative to the donors' natural variation ($3\sqrt{\lambda_d}$ standardized; 30 in whitened units, with a $\gamma = 10$ variant declared before its runs). τ is calibrated per regime so that a discriminant on the eight cue projections matches the centre-only discriminant on training-side data (0.969 high, 0.948 low; the centre-only benchmark on validation patients is 0.959 and 0.884); through a random encoder the encoded patch, centre included, reads at 0.94–0.99. The model is the linear encoder over the nine spectra and a ReLU head of width M over the encoded patch, trained and thresholded as in Section 4; three seeds per arm.

Results: high accessibility. The probe is 0.91–0.94 before training and changes little at L^* in any arm (gain -0.015 to $+0.035$): the accessible spectral information remains accessible, yet the informative-context-trained heads are strongly context-reliant, with reversal accuracy 0.08–0.10 and context-random accuracy at chance for widths 8–512 and for $\kappa = 1, 1/16, 1/256$ alike, all fitting within 70–230 steps; the uninformative-context comparator, also jointly trained, reaches reversal 0.93 and context-random 0.92, the frozen encoder 0.09 (Table 15). A large deficit in this probe is not necessary for failure.

Results: low accessibility. The probe starts at 0.56–0.60. Among the $\gamma = 30$ informative-context arms the mean gain at L^* is at most 0.046, with substantial seed variation (up to 0.10 in one run), against 0.35 for the uninformative-context comparator (probe 0.92–0.93), which needs 15,000–17,000 steps to reach L^* where the informative arms need 59–226; reversal accuracy is 0.07 against the comparator's 0.91–0.93, the frozen encoder giving 0.08 (Table 13). Slowing the head over the tested range gives only a small probe response (mean change $+0.014$ from $\kappa = 1$ to $1/256$) and does not restore the comparator's accessibility. The saved checkpoints show where the adaptation went: for every seed and κ, 99.995–99.998% of the squared net encoder update lies in the span of the eight constructed-cue directions, against 88–90% under uninformative-context training (Appendix D). That the cue's amplitude lets it dominate the encoder's own gradient is consistent with this and is offered as a supported hypothesis, not an identified mechanism: the projection is not a pathwise gradient decomposition, and the random head, unlike the theorem's zero-sum readout, lets contextual learning proceed through the encoder even when the head barely moves.

Results: the lower-amplitude variant. At $\gamma = 10$ the fit takes 530–1,800 steps and the rate response returns: the within-seed change in probe accuracy from $\kappa = 1$ to $1/256$ is $+0.030$, $+0.049$, $+0.095$, against $+0.002$, $+0.023$, $+0.016$ at $\gamma = 30$ in the same seeds (mean rate effect $+0.058$ against $+0.014$; the paired difference is positive in every seed), while reversal accuracy stays at 0.06–0.08 and the comparator again reaches a gain of 0.35 and reversal 0.92 (Table 14). The net update remains concentrated in the cue span (99.96–99.98%), so the variant shows an amplitude-dependent accessibility response, not the removal of contextual dominance.

Reading. With measured spectra the outcome matches the synthetic experiment on reliance: at every width and rate evaluated the original predictor uses the constructed context when it is informative, with excess shifted risk of 0.83–0.84 over the uninformative-context comparator at matched widths on validation patients. On adaptation it adds a scope condition: how much of the contrast the encoder learns before the fit depends on the head's relative speed and on the cue's amplitude, which are not interchangeable here; whether both act through the theorem's single parameter is a hypothesis these data do not decide.

6 Limitations and conclusion

Conclusion. We prove, in a serial linear-encoder model, that a contextual pathway trained faster than the spectral encoder suppresses the encoder's update at matched fit, that the fitted model fails under contextual reversal under explicit finite-width conditions, and that readout normalization removes the width dependence without necessarily removing the failure. At matched training loss in the synthetic spectral–spatial model, slowing the spatial head substantially increases the linear accessibility of the spectral cue while the trained classifier remains strongly context-reliant; retraining

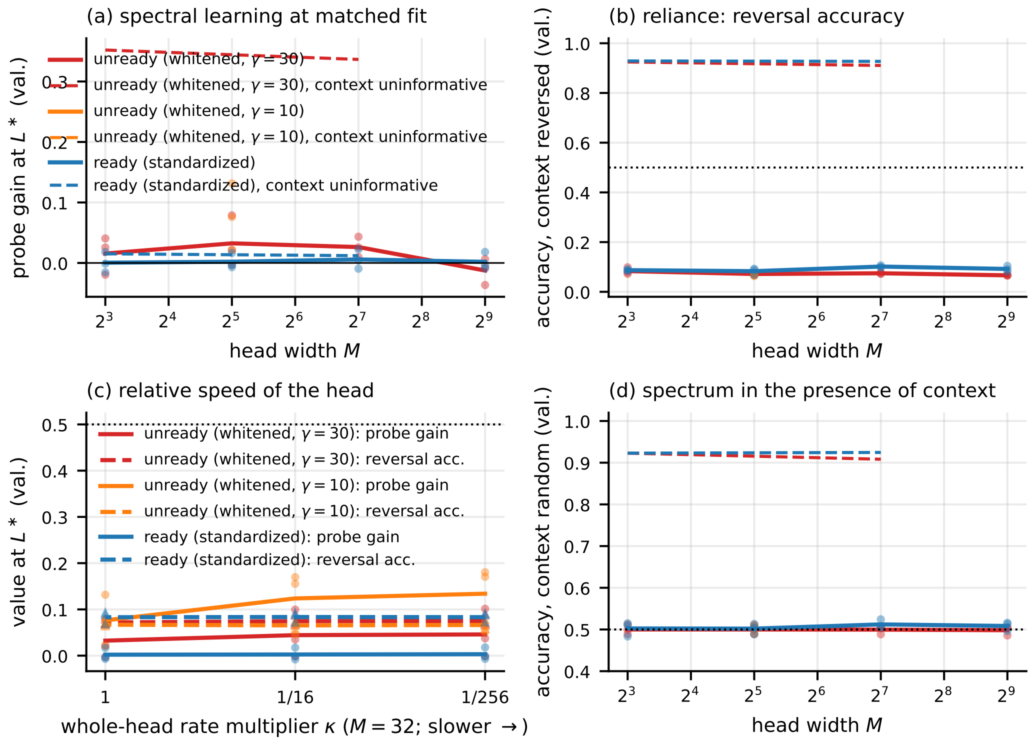

Figure 4: Experiment 3 at $L^* = 0.30$ on validation patients, three seeds per cell. (a) Probe gain against width: high accessibility (blue), low accessibility ($\gamma = 30$ red, $\gamma = 10$ orange), uninformative-context comparator dashed. (b) Accuracy under context reversal. (c) Whole-head rate multiplier at $M = 32$: probe gain (solid), reversal accuracy (dashed). (d) Accuracy with an uninformative context.

a fresh head on decorrelated context from the frozen encoders recovers about 0.2 more shifted accuracy after the slow head than after the fast one, from which no more is recovered than from a random encoder. With measured spectra, high spectral accessibility coexists with severe contextual reliance, so a deficit in this probe is not necessary for failure; with low initial accessibility, slowing the head yields only small accessibility gains at the larger constructed amplitude and a larger response at the smaller one. The four measurements respond differently to the same interventions; the paper's claim is that difference, at the stated fits, rates and constructions, not a general law.

Limitations. The theorem has a linear encoder, noiseless cues and a fixed spectral skip coefficient; the experiments add noisy cues and jointly learned routing but keep a linear encoder and one head family each, trained by plain gradient descent with a base rate and declared block multipliers. The contextual cues are constructed, so neither experiment shows that natural tissue neighbourhoods produce the effect, and matched mean training loss is an analogue of the theorem's matched margin, not the same rule. The accessibility asymmetry the mechanism needs is set by preprocessing and bottleneck: where a random encoder already reads the spectral cue, there is little for the encoder to gain, and the failure that still occurs there is not the theorem's. Real-spectrum seed variation is conditional on one patient split.

What the production model showed and did not. On a production segmentation model, initialization curvature is not the diagnostic used here: the scalar block ratio is inverted, coexisting with effectively rank-one inputs, its width trend is sensitive to the head's BatchNorm statistics, and the class contrast carries 2.6–$4.8\times$ less curvature than the dominant input direction, none of which measures impaired learning. A matched retraining study at width 48 and three seeds found an earlier apparent freezing advantage to be protocol-confounded and unresolved once matched (Appendix E).

We recommend no mitigation; whether the mechanism operates in production pipelines trained from scratch with natural context statistics remains open.

AI USE STATEMENT

Generative AI tools (large language model assistants) were used throughout this work: for literature search; for drafting and editing text; for writing and checking code, including the numerical verification scripts of Appendix F; and for adversarial review of the proofs and of the claims made about the measurements. Every proof was independently re-derived by the authors, and every number reported in the paper was recomputed by the authors from the recorded artifacts. All AI-generated text and code were reviewed by the authors, who take full responsibility for the content of this paper, including any remaining errors.

REPRODUCIBILITY STATEMENT

Code, configurations and audit scripts, with seeds and exact commands, are provided in the anonymized repository accompanying this submission. Every number in the paper traces to a named results file; Appendix F lists the verification scripts and their outcomes, including the two that do not exit cleanly. Complete hypotheses and proofs are in Appendix A, with the numerical verification in Appendix B. Generators, calibration, architectures, data, patient splits and training protocols are described in Appendices C, D and E.

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A PROOFS OF THEOREM 1

Theorem 1 combines three statements proved below in their original form: the serial instance at the special initialization (Theorem A.1), its general and isotropic initialization (Corollary A.1), and the readout normalization (Proposition A.2). Part (a) of Theorem 1 is (10) together with the finite-horizon gap (12); part (b) is the reversal statement of Corollary A.1 including (13); part (c) is Proposition A.2 with $\alpha_M = M^{-1/2}$, whose proof applies to any zero-sum initialization of the normalized readouts because only the aggregate b enters the flow. The rate-κ variant quoted in Section 3 follows by the same differentiation with $\dot{\beta}_j = \kappa v \varrho$. The special-initialization remark is (4) and the GGN display of Theorem A.1. Notation: $\varrho = (1 + e^q)^{-1}$ is the logistic residual, $\Psi(q) = q + e^q - 1$, and $G_{\boldsymbol{\theta}\boldsymbol{\theta}}$, $G_{\boldsymbol{\phi}\boldsymbol{\phi}}$ are the Gauss–Newton blocks of the scalar-logit loss in the parameter blocks $\boldsymbol{\theta} = (a, v)$ and $\boldsymbol{\phi} = \beta$.

A.0.1 THE SERIAL INSTANCE, ITS GENERAL INITIALIZATION, AND THE NORMALIZED READOUT

Proposition A.1 (Information retained by a fitting serial encoder)**.** *If the predictions depend on X only through $Z = f_{\boldsymbol{\theta}}(X)$ and a classifier $q_{\boldsymbol{\phi}}(Y \mid Z)$ has population CE at most $\eta < \infty$, then $H(Y \mid Z) \le \eta$ and $I(Y; Z) \ge H(Y) - \eta$. The same statement holds for the empirical distribution with a fixed model.*

Proof. CE is conditional entropy plus the nonnegative expected conditional KL divergence. Suppression of a selected coordinate or failure under a specified shift is therefore different from absence of all training-label information in the serial representation. $\square$

Theorem A.1 (Serial CE instance; review Theorem 5.1)**.** *Let Y be uniform on $\{-1, 1\}$ and let training inputs be $x_1 = (S, 0)$, $x_2 = (0, C_{\text{ctx}})$ with $S = C_{\text{ctx}} = Y$. The shared linear encoder is $W = (a, v)$, giving $z_1 = aS, z_2 = vC_{\text{ctx}}$. For integer $M \ge 1$ define*

$$F = z_1 + \sum_{j=1}^{M} \beta_j z_2 = aS + bvC_{\text{ctx}}, \quad b = \sum_j \beta_j, \quad \boldsymbol{\theta} = (a, v), \quad \boldsymbol{\phi} = \beta.$$

All coordinates follow unit Euclidean gradient flow on the averaged binary CE $\mathcal{L} = \mathbb{E} \log(1 + e^{-YF}) = \log(1 + e^{-q})$, $q = a + bv$, from $a_0 = 0, v_0 = 1, \beta_0 = 0$, with no clipping, decay or momentum. Fix $0 < \delta < 1/2$, $m = \log((1-\delta)/\delta)$, and let T_m be the first time q reaches m. Write $\varrho = (1 + e^q)^{-1}$ and $\Psi(q) = q + e^q - 1$. Then:

$$\dot{a} = \varrho, \qquad \dot{v} = b\varrho, \qquad \dot{b} = Mv\varrho, \tag{1}$$

$$v = \cosh(\sqrt{M} a), \qquad b = \sqrt{M} \sinh(\sqrt{M} a), \tag{2}$$

$$q = a + \frac{\sqrt{M}}{2} \sinh(2\sqrt{M} a), \qquad \dot{q} = (M + 1 + 2b^2)\varrho. \tag{3}$$

The solution is global and reaches m in finite time, with

$$\frac{\Psi(m)}{M + 1 + 2m^2} \le T_m \le \frac{\Psi(m)}{M + 1}. \tag{4}$$

At that time, writing $a_M = a(T_m)$,

$$0 < a_M \le \frac{m}{M + 1}, \qquad b(T_m)v(T_m) \ge \frac{Mm}{M + 1}, \qquad Ma_M \longrightarrow m, \tag{5}$$

$$0 \le v(T_m) - 1 \le \frac{m^2}{2M}, \qquad \|W(T_m) - W_0\| \le \sqrt{\frac{m^2}{(M + 1)^2} + \frac{m^4}{4M^2}}. \tag{6}$$

For fixed m, a_M decreases strictly with M; removing the contextual path entirely requires $a = m$ at the same fitting margin. Moreover,

$$\varrho(t) \le \frac{1/2}{1 + (M + 1)t/4}, \qquad t \ge 0. \tag{7}$$

In the stated parameter blocks and loss convention,

$$\lambda_{\max}(G_{\boldsymbol{\theta\theta}}) = \varrho(1 - \varrho)(1 + b^2), \quad \lambda_{\max}(G_{\boldsymbol{\phi\phi}}) = M\varrho(1 - \varrho)v^2,$$

so at initialization they equal $1/4, M/4$ and the disparity is M. For the separately trained frozen encoder $W = (0, 1)$, $\beta_F(0) = 0$, its margin obeys $\Psi(q_F(t)) = Mt$ and

$$0 \le q(t) - q_F(t) \le \frac{(1 + 2m^2)t}{2 + Mt/2} \le \frac{2(1 + 2m^2)}{M}, \qquad 0 \le t \le T_m. \tag{8}$$

Under the specified reversal $S = Y, C_{\text{ctx}} = -Y$, the joint model at T_m has error one; the fitted spectral-only comparator has error zero.

Proof. The training margin is the same on each example, so direct differentiation gives (1). Divide by $\dot{a} = \varrho > 0$ to get $dv/da = b$, $db/da = Mv$, which proves (2). It implies $b^2 = M(v^2 - 1)$ and (3). Since $0 < \dot{a} \leq 1/2$, a stays finite on every finite time interval; the phase formula then gives global existence. The strictly increasing map $Q_M(a) = a + (\sqrt{M}/2)\sinh(2\sqrt{M}a)$ maps $[0, \infty)$ onto itself and has a finite inverse at m. On this compact a interval, $\dot{a} \geq \delta > 0$, so the hitting time is finite.

$\sinh x \geq x$ gives $Q_M(a) \geq (M+1)a$, strictly for $a > 0$. Before fitting, $v \geq 1$, $0 \leq bv = q - a \leq m$, hence $b \leq m$. The invariant gives $v - 1 = b^2/[M(v+1)] \leq m^2/(2M)$. For fixed positive a the contextual term increases strictly with M; invert to get strict decrease of a_M. Its upper bound implies $\sqrt{M}a_M \to 0$, and the expansion $m = (M+1)a_M + O(M^2 a_M^3)$ gives $Ma_M \to m$.

$\dot{\varrho} = -(M + 1 + 2b^2)\varrho^2(1 - \varrho)$ and $\varrho \leq 1/2$ imply $(d/dt)(1/\varrho) \geq (M+1)/2$. Integrate from $\varrho(0) = 1/2$ for the envelope. Since $\Psi'(q) = 1/\varrho$, before fitting $\frac{d}{dt}\Psi(q) = M + 1 + 2b^2 \in [M+1, M+1+2m^2]$; integration proves (4). The margin gradients are $Y(1, b)$ and $Yv\mathbf{1}_M$; the scalar logistic Hessian is $\varrho(1 - \varrho)$, giving the GGN.

For the frozen system $\dot{q}_F = M/(1+e^{q_F})$. Thus $0 \leq \Psi(q) - \Psi(q_F) = \int_0^t (1+2b^2)ds \leq (1+2m^2)t$. Convexity gives a lower bound $\Psi'(q_F)(q - q_F)$ for this difference. As $q_F \leq e^{q_F} - 1$, $Mt = \Psi(q_F) \leq 2(e^{q_F} - 1)$, so $\Psi'(q_F) \geq 2 + Mt/2$, proving (8). The shifted margin is $a - bv = 2a_M - m < 0$: for $M > 1$ use (5), and for $M = 1$ use the strict inequality $Q_1(a) > 2a$. The spectral-only shifted margin is m. ☐

Remark A.1 (Embedding, head, and initialization conventions). For logits $(F, 0)$, loss, residual norm and GGN agree, but the unprojected logit-Jacobian norm is $\sqrt{1+b^2}$; the symmetric embedding makes the norm-product bound exact. A freely trained dense two-row head is not the parameter block $\boldsymbol{\phi} = \beta$: its initial head eigenvalue is $M/2$, not $M/4$, while the encoder eigenvalue remains $1/4$ at the same initial function. Labels need not be balanced for the training ODE; balance is used in the information/noisy-accuracy statement below. The initial encoder orientation in Theorem A.1 is explicitly asymmetric; the next corollary removes that choice for W, not the routing/readout asymmetry.

Proposition A.2 (Readout normalization and representation interpretation)**.** *Replace the contextual readout by $\alpha_M \sum_j \gamma_j z_2$, $\gamma_j(0) = 0$, at unit rates. Its reduced equation for $b = \alpha_M \sum_j \gamma_j$ is $\dot{b} = M\alpha_M^2 v\varrho$. In particular $\alpha_M = M^{-1/2}$ gives exactly the $M = 1$ dynamics for every M. At every finite M in the original theorem, the noiseless spectral coordinate $a_M S$ still determines Y. If it is instead observed as $Z_S = a_M Y + \sigma\xi$, with $\xi \sim \mathcal{N}(0, 1)$ independent, $\sigma > 0$ fixed, then Bayes accuracy is $\Phi(a_M/\sigma)$ and $I(Y; Z_S) \leq a_M^2/(2\sigma^2)$.*

Proof. Differentiate each readout and sum. The Gaussian likelihood-ratio test is thresholding at zero, giving the accuracy. For $Q = \mathcal{N}(0, \sigma^2)$, $\mathbb{E}_Y \operatorname{KL}(P_{Z_S|Y} \| Q) = a_M^2/(2\sigma^2) = I(Y; Z_S) + \operatorname{KL}(P_{Z_S} \| Q)$. This noise is a specified observation perturbation, not training noise. ☐

Corollary A.1 (General and isotropic encoder initialization)**.** *In the serial model let $m > 0$, $a_0 < m$, $v_0 \neq 0$, and allow any readout initialization with $\sum_j \beta_{j0} = 0$. Set $u = a - a_0$, $\Delta = m - a_0 > 0$. Then*

$$v = v_0 \cosh(\sqrt{M}u), \quad b = \sqrt{M}v_0 \sinh(\sqrt{M}u), \tag{9}$$

$$q = a_0 + u + \frac{\sqrt{M}v_0^2}{2}\sinh(2\sqrt{M}u),$$

$$0 < u_M \leq \frac{\Delta}{1 + Mv_0^2}, \qquad Mu_M \longrightarrow \frac{\Delta}{v_0^2}, \qquad \frac{a(T_m) - a_0}{m - a_0} \longrightarrow 0. \tag{10}$$

The fitting root decreases strictly with M and

$$|v(T_m) - v_0| \leq \frac{\Delta^2}{2M|v_0|^3}, \qquad \|W(T_m) - W_0\| \leq \sqrt{\frac{\Delta^2}{(1 + Mv_0^2)^2} + \frac{\Delta^4}{4M^2|v_0|^6}}. \tag{11}$$

With $\Psi_{a_0}(q) = q - a_0 + e^q - e^{a_0}$,

$$\frac{\Psi_{a_0}(m)}{1 + Mv_0^2 + 2\Delta^2/v_0^2} \leq T_m \leq \frac{\Psi_{a_0}(m)}{1 + Mv_0^2}.$$

For a frozen encoder at (a_0, v_0) *and the same initial readouts,* $\Psi_{a_0}(q_F) = Mv_0^2 t$*. If* $p_0 = e^{a_0}/(1 + e^{a_0})$ *and* $C_0 = 1 + 2\Delta^2/v_0^2$*, then*

$$0 \le q_J - q_F \le \frac{C_0 t}{1 + e^{a_0} + p_0 M v_0^2 t} \le \frac{C_0}{p_0 M v_0^2}, \qquad 0 \le t \le T_m. \tag{12}$$

Under contextual reversal the fitted margin is $2a(T_m) - m$*. For* $a_0 < m/2$*, error is one for all sufficiently large M, with sufficient condition* $Mv_0^2 > m/(m - 2a_0)$*. For* $m/2 \le a_0 < m$*, the reversed classifier is correct at every width. If* $a_0 \ge m$*, define fitting to stop immediately, in which case it is also correct.*

If $(a_0, v_0) \sim \mathcal{N}(0, \sigma^2 I_2)$*,* $\sigma > 0$*, then* $v_0 \ne 0$ *almost surely; use the preceding stopping convention. The expected reversal error satisfies*

$$\lim_{M\to\infty} \mathbb{E}_{W_0} \operatorname{Err}_{\text{reversal}} = \Phi\left(\frac{m}{2\sigma}\right). \tag{13}$$

The same-threshold spectral-only comparator has zero reversal error.

Proof. Dividing the ODE by $\dot{a} > 0$ gives (9); only the initial readout sum enters. Each $\beta_i - \beta_j$ is conserved, so equal initial readouts are not needed for closure. Positivity of the contextual margin $bv = (\sqrt{M} v_0^2/2)\sinh(2\sqrt{M}u)$ and the sinh inequality prove (10); Taylor expansion at $\sqrt{M}u_M \to 0$ gives the limit. The invariant is $b^2 = M(v^2 - v_0^2)$; v keeps its sign and $|v| \ge |v_0|$. Since $bv \le \Delta$, $|b| \le \Delta/|v_0|$ and

$$|v - v_0| = \frac{b^2}{M(|v| + |v_0|)} \le \frac{\Delta^2}{2M|v_0|^3}.$$

This finite-M inequality proves (11). Now $\dot{q} = (1 + Mv_0^2 + 2b^2)\varrho$ and $b^2 \le \Delta^2/v_0^2$ on the fitting interval; integrate the transformed margin for the time bounds. Global existence follows as before from $\dot{u} \le 1$ and the phase formula; every finite m is hit.

The transformed joint/frozen difference lies between zero and $C_0 t$. Also $q_F - a_0 \le (e^{q_F} - e^{a_0})/e^{a_0}$ gives $\Psi'_{a_0}(q_F) \ge 1 + e^{a_0} + p_0 M v_0^2 t$. Convexity proves (12). For $a_0 < m/2$, use $a(T_m) \le a_0 + \Delta/(1 + Mv_0^2)$ to get the finite-M reversal condition. For $a_0 > m/2$ the classifier is correct at every fitted width, since $u_M > 0$ gives $2a(T_m) - m > 2a_0 - m > 0$; equality $a_0 = m/2$ also gives a positive margin. Immediate stopping covers $a_0 \ge m$. Almost surely the limiting error is $1_{\{a_0 < m/2\}}$, apart from the zero-probability boundary and $v_0 = 0$ events. Dominated convergence gives (13). □

Remark A.2 (Confidence dependence and limiting quantity). For $a_0 \ne 0$, $a(T_m)/a_0 \to 1$ but $a(T_m)/m \to a_0/m$; the vanishing quantity is the *update* fraction in (10). For $0 < \rho < 1$, a Gaussian tail and its maximum density give, with probability at least $1 - \rho$,

$$|a_0| \le \sigma\sqrt{2\log(4/\rho)}, \qquad |v_0| \ge \frac{\rho\sigma}{2}\sqrt{\pi/2}.$$

Substitute these bounds in the finite-M inequalities for uniform confidence bounds. The dependence is severe: for fixed $(\Delta, v_0) \in (0, \infty) \times (\mathbb{R} \setminus \{0\})$, expansion of the root yields

$$Mu_M = \frac{\Delta}{v_0^2}\left[1 - \frac{1}{Mv_0^2} - \frac{2\Delta^2}{3Mv_0^4} + O(M^{-2})\right].$$

Indeed substitute $u = A/M + B/M^2$ in the sinh equation and match its constant and M^{-1} coefficients. The expansion's remainder is pointwise, not already uniform in rho. Its small-$|v_0|$ crossover requires control of $\Delta^2/(Mv_0^4)$; at the confidence cutoff this has scale $\Delta^2\rho^{-4}\sigma^{-4}/M$, a fourth power of $1/\rho$, not a second. A rigorous uniform remainder can instead be bounded from the sinh series on this controlled argument. Do not present this asymptotic crossover as a universal necessary width.

Provenance and interpretation. Architecture- and initialization-dependent implicit bias is established in linear-network analyses (Yun et al., 2021; Moroshko et al., 2020); trajectory-level incremental learning is studied by Berthier (2023). Exact deep-linear dynamics (Saxe et al., 2014) and conservation of layer-norm differences (Du et al., 2018) supply the underlying tools. CE feature competition is central to Pezeshki et al. (2021). The serial example instantiates a known optimization-metric mechanism with finite-margin formulas, an encoder-update comparison and a specified reversal distribution. Replication is equivalent to a rate M on b, and normalization removes it; the function class does not grow. These particular calculations are not a claim of a new general implicit-bias mechanism, and finite-time or serial composition alone is not a novelty claim.

B EXACT-MODEL VERIFICATION

The closed forms of Appendix A were checked against a full $(2+M)$-parameter integration of the gradient flow. The tables below are reproduced from the verification record of the earlier draft (`code/experiments/toy_serial_ce.py`, `toy_serial_ce_isotropic.py`); the maximum relative discrepancy over the four endpoint quantities in the 525 non-trivial isotropic cases is 1.26×10^{-11}, the maximum trajectory-identity discrepancy is 1.5×10^{-10}, and no bound is violated in any case. The quantities and error conventions are those stated in each caption.

B.0.1 SOLVABLE SERIAL INSTANCE: NUMERICAL VERIFICATION

Zero-context instance. Table 1 verifies the closed forms and the bounds of the solvable serial cross-entropy instance at $a_0 = 0$, $v_0 = 1$ and margin $m = \log 19 = 2.944$, from `results/toy_serial_ce.csv`. The solver column is a stiff ODE integration of the full $(2+M)$-dimensional state; a separate explicit-Euler gradient-descent path, run at four step sizes, has observed convergence order 1.0000004 and agrees with an independent numpy analytic-gradient implementation to 0 relative error.

Table 1: Solvable serial instance ($a_0 = 0$, $v_0 = 1$, $m = \log 19$): closed form against ODE solver, from `results/toy_serial_ce.csv`. The displacement block compares the realised encoder displacement with the displacement bound of the earlier draft's framework, which is not part of this paper's theory and is retained for the record only. The relative discrepancies between closed form and solver for the endpoint quantities in this table are at most 1.2×10^{-13}; the trajectory-invariant check of the next table reaches 2.0×10^{-11}.

Quantity	$M = 16$	$M = 64$	$M = 256$	$M = 1024$
$\lambda_{\boldsymbol{\theta}}(0)$	0.25	0.25	0.25	0.25
$\lambda_{\boldsymbol{\phi}}(0)$	4.0	16.0	64.0	256.0
$\mathcal{D}_{\mathrm{curv}}(0)$	16.0	64.0	256.0	1024.0
$a(T_m)$	0.142325	0.042094	0.011216	0.002857
suppression m/a_M	20.688	69.949	262.53	1030.7
$M\,a_M$ (limit m)	2.2772	2.6940	2.8712	2.9253
T_m	0.90784	0.28680	0.078763	0.020252
T_m lower bound $\Psi(m)/(M+1+2m^2)$	0.60992	0.25437	0.076345	0.020094
T_m upper bound $\Psi(m)/(M+1)$	1.2320	0.32222	0.081496	0.020434
$\sup_{0\le t\le T_m}(q_J - q_F)$	0.38232	0.13813	0.039802	0.010376
uniform gap bound	11.297	2.9547	0.74729	0.18737
$(M+1)\times$ gap	6.4994	8.9787	10.229	10.635
$\Vert W(T_m) - W(0)\Vert_2$	0.219021	0.071050	0.019658	0.005064
displacement bound	1.370601	0.358465	0.090662	0.022732
bound / displacement	6.258	5.045	4.612	4.489
normalized-readout control: $a(T_m)$	1.026947	1.026947	1.026947	1.026947
normalized-readout control: $\Vert W(T_m)-W(0)\Vert$	1.177117	1.177117	1.177117	1.177117
residual envelope excess on $[0, 20T_m]$	0	0	0	0
readout spread $\max_j \beta_j - \min_j \beta_j$ at T_m	0	0	$1.4 \cdot 10^{-17}$	0
invariant $q = Q_M(a)$, max rel. error	$2.0 \cdot 10^{-11}$	$6.7 \cdot 10^{-13}$	$1.5 \cdot 10^{-13}$	$6.8 \cdot 10^{-14}$

The bound stays within 4.5 to 6.3 times the realised displacement across this width range while both fall by more than a factor of 40. The normalized-readout control, in which the readout is $M^{-1/2}Uh$ with $O(1)$ entries trained at unit rate, removes the width dependence exactly: the encoder's movement is the $M = 1$ value at every width, to 1.7×10^{-14} relative error. The eigenvalue constants $\lambda_{\boldsymbol{\theta}}(0) = 1/4$ and $\lambda_{\boldsymbol{\phi}}(0) = M/4$ hold for the mean-over-examples binary cross-entropy with a unit-scaled scalar logit; summing instead of averaging, or rescaling the logit, multiplies both blocks equally and leaves $\mathcal{D}_{\mathrm{curv}}(0) = M$ unchanged.

Isotropic extension. Table 2 collects the verification of the general-initialization version, in which $a(0) = a_0$, $v(0) = v_0 \neq 0$ and the readouts start unequal but sum to exactly zero. The grid is $M \in$

$\{1, 4, 16, 64, 256, 1024, 4096\}$, $a_0 \in \{-1.5, -0.3, 0, 0.4, 1.7\}$, $v_0 \in \{-1.3, -0.2, 0.05, 0.2, 1.0\}$ and $m \in \{2.0, 2.9, 4.0\}$: 525 non-trivial cases plus 54 immediate-stop cases with $a_0 \geq m$, integrated with DOP853 at `rtol` 10^{-10} and `atol` 10^{-12} with a terminal event at $q = m$.

Table 2: Isotropic serial instance: verification summary and worst-case bound slacks over the whole grid, from `results/toy_serial_ce_isotropic_REPORT.md` §1–§2. Slacks are bound − value for upper bounds and value − bound for lower bounds; a negative entry would be a violation, and there are none. $\Delta = m - a_0$.

Quantity	Value
Max closed-form vs ODE relative error over u, $v(T_m)$, $b(T_m)$, T_m (525 cases)	1.260×10^{-11}
Max relative error of the conserved quantity along the whole trajectory	1.518×10^{-10}
Max root accuracy $\lvert Q_{\mathrm{iso}}(u_M) - m\rvert$	3.553×10^{-15}
Max $\lvert \sum_j \beta_j(0)\rvert$ (zero-sum initialization is bitwise exact)	0
Max spread of $\beta_j(T_m) - \beta_j(0)$	7.772×10^{-16}
Max frozen-comparator ODE vs closed-form relative error	5.993×10^{-11}
Max $M^{-1/2}$-readout control vs its $M = 1$ value	1.335×10^{-12}
Bound violations, all cases and all bounds	0
Min slack, $u_M \leq \Delta/(1 + Mv_0^2)$	2.221×10^{-10}
Min slack, $u_M > 0$	4.333×10^{-5}
Min slack, $\lvert v(T_m) - v_0\rvert \leq \Delta^2/(2M\lvert v_0\rvert^3)$	1.493×10^{-9}
Min slack, encoder displacement bound	3.918×10^{-10}
Min slack, $T_m \leq \Psi_{a_0}(m)/(1 + Mv_0^2)$	1.748×10^{-9}
Min slack, $T_m \geq \Psi_{a_0}(m)/(1 + Mv_0^2 + 2\Delta^2/v_0^2)$	3.175×10^{-9}
Min slack, $q_J(t) \geq q_F(t)$	0
Min slack, $\sup_t(q_J - q_F) \leq C_0/(p_0 M v_0^2)$	1.495×10^{-4}
Reversal error at $\sigma = 1$, $m = 2$, $M = 64$ / 1024 / 16384	0.8207 / 0.8380 / 0.8397
Monte-Carlo limit $\Phi(m/2\sigma)$ on the same 4000 draws	0.8413
Draws disagreeing with the limit that satisfy $Mv_0^2 > m/(m - 2a_0)$	0 of 48000
Spectral-only comparator reversal error, every case	0

The update fraction $(a(T_m) - a_0)/(m - a_0)$ falls towards zero in M at every (a_0, v_0) on the grid — for instance 0.288, 0.0423, 0.000964 at $M = 1, 16, 1024$ with $a_0 = -1.5$, $v_0 = 1$, $m = 2.9$ — while $a(T_m)/m$ and $a(T_m)/a_0$ converge to a_0/m and to 1 instead, so those two ratios are not the quantities that vanish. Under the $M^{-1/2}$ readout the suppression is 0.336399 at every width, against 0.336, 0.0472 and 0.000969 for the unnormalized readout at $M = 1$, 16 and 1024. Every Monte-Carlo draw that falls short of the reversal limit fails the sufficient finite-width condition, either because $\lvert v_0\rvert$ is small or because a_0 sits just below $m/2$, where the threshold diverges.

C Experiment 2 details

C.1 Generator

Images are 16×16 on a torus with $S = 256$ channels per pixel. Labels $y_p \in \{-1, +1\}$ are i.i.d. Rademacher. Fixed orthonormal directions $u, v_1, \ldots, v_8 \in \mathbb{R}^S$ are drawn once per seed (QR of a Gaussian matrix). With $o_1, \ldots, o_8$ the eight 3×3 offsets, the spectrum of pixel p is

$$x_p = \alpha\, y_p\, u + \beta \sum_{d=1}^{8} \big(y_{p-o_d} + \tau\, \eta_{p,d}\big) v_d + \sigma\, \varepsilon_p, \qquad \eta_{p,d}, \varepsilon_p \text{ standard Gaussian,}$$

so that the pixel at $p + o_d$ carries y_p along v_d: each pixel's label is written into all eight neighbours, each along its own direction, while p's own v-coordinates carry the labels of p's neighbours, which are independent of y_p. Constants: $\alpha = 1.645$, $\beta = 5$, $\sigma = 1$, $\tau = 1.7023$; $K = 12$. The draw order is condition-independent, so the five conditions generated from one seed share y, η and ε: *iid*; *reversed* ($y_{p-o_d} \to -y_{p-o_d}$); *context-random* (an independent label field replaces y in the context term); *spectral-only* ($\beta = 0$); *context-only* ($\alpha = 0$). Training uses 256 images; each test condition 64 images; the spectral probe is fitted on 64 spectral-only images and evaluated on another 64. Seeds:

directions $1000 + s$, training data $2000 + s$, test family $3000 + s$, probe $4000 + s$ and $5000 + s$, initialization s, for $s \in \{0, 1, 2\}$.

C.2 Calibration and readiness (fixed before any training run)

The oracle spectral accuracy is a Fisher discriminant on $u^\top x_p$; the oracle contextual accuracy is a Fisher discriminant on the 3×3 window of the eight projections $v_d^\top x$ (72 features), which contains all eight copies of y_p. τ was set by bisection so that the contextual oracle equals 0.950 on seed-0 directions. Readiness is the same discriminant through a random encoder $W \sim \mathcal{N}(0, 1/S)^{12\times256}$: on the encoder output of the pixel itself for spectral-only data, and on the 3×3 window of encoder outputs for context-only data.

Table 3: Experiment 2 calibration (`results/exp2/calibration.json`). Oracle accuracies use the true directions; readiness uses a random encoder.

seed	oracle spectral	oracle context	random enc. spectral	random enc. context	random enc. both
0	0.951	0.950	0.647	0.910	0.915
1	0.951	0.949	0.653	0.914	0.918
2	0.948	0.950	0.608	0.905	0.906

The population optimum of the contextual cue alone is a cross-entropy of about 0.126 nats (spectral alone 0.127); the threshold 0.15 sits just above that population optimum. A population optimum is not a bound on the finite-sample training loss, which a wide head can drive lower by fitting individual pixels, so training losses below 0.13 are not by themselves evidence of spectral learning; the probe and the shifted accuracies are.

C.3 Model, training and arms

Encoder $W \in \mathbb{R}^{12\times256}$, no bias, $\mathcal{N}(0, 1/S)$ initialization. Head: 3×3 convolution to width M with bias, ReLU, 3×3 convolution to one logit without bias, circular padding, PyTorch default initialization; implemented as explicit matrix products with torus rolls (identical to the convolution to 10^{-15} in double precision). Loss: mean over the 65,536 training pixels of $\log(1 + e^{-y_p F_p})$. Full-batch gradient descent, $\eta = 10^{-3}$ for every parameter (the largest of $\{10^{-4}, 3\cdot10^{-4}, 10^{-3}, 3\cdot10^{-3}\}$ with a monotone loss over 300 steps at $M = 2048$), no momentum, clipping or decay; at most 40,000 steps, stopping at loss < 0.10. Snapshots at the first step with loss below each of $0.6, 0.5, 0.4, 0.3, 0.2, 0.15, 0.10$; $L^* = 0.30$ is the pre-registered threshold. Gradient norms at a snapshot are those of the snapshotted iterate. Arms: `sp` (standard parameterization) and `mup` (readout $M^{-1/2}\gamma$, $\gamma = \sqrt{M}\times$default, same function at initialization; under plain gradient descent equal to a readout-only rate $1/M$) at $M \in \{2, 4, 8, 32, 128, 512, 2048\}$; `ctxfree` (training context from an independent label field) and `frozen` (encoder fixed at initialization) at $M \in \{2, 8, 128, 512\}$; `lrmult` (readout rate $\times\{1/16, 1/4, 1, 4, 16\}$) and `headlr` (whole-head rate $\kappa \in \{1, 1/16, 1/256\}$, extended to $1/4096$ and $1/32768$ on seed 0 for the reversal outcome) at $M = 32$. Three seeds.

C.4 Pre-registration and amendments

The plan (`paper/REFOCUS_PLAN_2026-09-10.md`, §4) fixed the generator, constants, arms, L^*, measures and predictions P1–P5 before the grid. Amendments recorded before the corresponding runs: widths 2 and 4 and fractional readout multipliers were added after a one-seed pilot at $M \in \{8, 2048\}$ showed complete suppression at L^* (§4.4a); the whole-head arm, the exact h_u export, paired test conditions and same-state gradient norms were added after the design review (§4.4b); the κ bracket was extended downward after the seed-0 pilot showed that the probe responds to κ while reversal accuracy does not within $\{1, 1/16, 1/256\}$ (§4.4c). Original predictions are reported as registered; added widths and multipliers are identified as amendments.

C.5 RESULTS

Tables generated by `paper/figures_src/make_tables_exp2.py` from `results/exp2_summary.csv`; the registered verdicts are the per-seed Spearman correlations listed at the end.

Table 4: Experiment 2 at the pre-registered threshold $L^* = 0.30$: mean±sd over seeds (number of seeds). Probe gain is the discriminant accuracy on the encoder output minus its initial value (0.647, 0.653, 0.608 for seeds 0–2); $h_u = \|P_{\mathrm{row}(W)} u\|^2$ (0.055 at initialization).

arm	M	probe gain	h_u	reversed acc.	context-random acc.	spectral-only acc.
`sp`	2	0.042±0.028 (3)	0.086±0.037 (3)	0.134±0.006 (3)	0.503±0.005 (3)	0.525±0.016 (3)
`sp`	4	0.021±0.010 (3)	0.065±0.028 (3)	0.125±0.008 (3)	0.508±0.007 (3)	0.525±0.021 (3)
`sp`	8	0.026±0.016 (3)	0.067±0.021 (3)	0.124±0.002 (3)	0.507±0.005 (3)	0.524±0.017 (3)
`sp`	32	0.022±0.007 (3)	0.067±0.027 (3)	0.121±0.008 (3)	0.508±0.008 (3)	0.531±0.022 (3)
`sp`	128	0.009±0.006 (3)	0.055±0.022 (3)	0.117±0.012 (3)	0.507±0.008 (3)	0.524±0.019 (3)
`sp`	512	0.005±0.001 (3)	0.051±0.018 (3)	0.115±0.007 (3)	0.508±0.006 (3)	0.525±0.022 (3)
`sp`	2048	0.002±0.000 (3)	0.049±0.017 (3)	0.117±0.005 (3)	0.507±0.006 (3)	0.522±0.017 (3)
`mup`	2	0.054±0.034 (3)	0.098±0.042 (3)	0.134±0.006 (3)	0.504±0.004 (3)	0.528±0.015 (3)
`mup`	4	0.036±0.019 (3)	0.079±0.037 (3)	0.126±0.011 (3)	0.507±0.007 (3)	0.530±0.026 (3)
`mup`	8	0.039±0.023 (3)	0.080±0.034 (3)	0.120±0.002 (3)	0.506±0.006 (3)	0.530±0.024 (3)
`mup`	32	0.044±0.014 (3)	0.088±0.037 (3)	0.115±0.009 (3)	0.507±0.007 (3)	0.539±0.026 (3)
`mup`	128	0.026±0.016 (3)	0.070±0.031 (3)	0.108±0.012 (3)	0.505±0.008 (3)	0.526±0.018 (3)
`mup`	512	0.026±0.019 (3)	0.071±0.034 (3)	0.112±0.008 (3)	0.505±0.007 (3)	0.526±0.022 (3)
`mup`	2048	0.028±0.004 (3)	0.070±0.018 (3)	0.104±0.005 (3)	0.504±0.006 (3)	0.523±0.010 (3)
`ctxfree`	2	0.296±0.019 (3)	0.863±0.048 (3)	0.877±0.002 (3)	0.876±0.003 (3)	0.877±0.002 (3)
`ctxfree`	8	0.281±0.022 (3)	0.741±0.014 (3)	0.893±0.001 (3)	0.893±0.001 (3)	0.899±0.002 (3)
`ctxfree`	128	0.268±0.023 (3)	0.650±0.014 (3)	0.896±0.001 (3)	0.895±0.002 (3)	0.903±0.004 (3)
`ctxfree`	512	0.253±0.025 (3)	0.564±0.005 (3)	0.886±0.002 (3)	0.887±0.004 (3)	0.891±0.004 (3)
`frozen`	2	0.000±0.000 (3)	0.048±0.017 (3)	0.139±0.009 (3)	0.508±0.006 (3)	0.521±0.018 (3)
`frozen`	8	0.000±0.000 (3)	0.048±0.017 (3)	0.134±0.006 (3)	0.511±0.007 (3)	0.524±0.019 (3)
`frozen`	128	0.000±0.000 (3)	0.048±0.017 (3)	0.127±0.008 (3)	0.508±0.008 (3)	0.524±0.021 (3)
`frozen`	512	0.000±0.000 (3)	0.048±0.017 (3)	0.120±0.006 (3)	0.508±0.008 (3)	0.526±0.019 (3)

Table 5: Experiment 2 at the secondary threshold 0.15 (labelled; not pre-registered): mean±sd over seeds that reached it.

arm	M	probe gain	h_u	reversed acc.	context-random acc.
`sp`	2	0.219±0.039 (3)	0.462±0.232 (3)	0.131±0.088 (3)	0.543±0.046 (3)
`sp`	4	0.094±0.019 (3)	0.149±0.056 (3)	0.070±0.006 (3)	0.506±0.007 (3)
`sp`	8	0.121±0.020 (3)	0.184±0.046 (3)	0.073±0.007 (3)	0.508±0.008 (3)
`sp`	32	0.094±0.014 (3)	0.147±0.051 (3)	0.072±0.008 (3)	0.509±0.009 (3)
`sp`	128	0.060±0.022 (3)	0.106±0.052 (3)	0.070±0.009 (3)	0.507±0.009 (3)
`sp`	512	0.041±0.012 (3)	0.084±0.036 (3)	0.071±0.011 (3)	0.508±0.009 (3)
`sp`	2048	0.032±0.005 (3)	0.073±0.026 (3)	0.076±0.010 (3)	0.509±0.010 (3)
`mup`	2	0.230±0.032 (3)	0.499±0.216 (3)	0.136±0.093 (3)	0.546±0.051 (3)
`mup`	4	0.142±0.020 (3)	0.233±0.079 (3)	0.074±0.008 (3)	0.508±0.006 (3)
`mup`	8	0.148±0.030 (3)	0.241±0.090 (3)	0.073±0.009 (3)	0.509±0.008 (3)
`mup`	32	0.134±0.015 (3)	0.213±0.068 (3)	0.070±0.008 (3)	0.507±0.008 (3)
`mup`	128	0.088±0.033 (3)	0.146±0.070 (3)	0.064±0.006 (3)	0.505±0.008 (3)
`mup`	512	0.108±0.041 (3)	0.178±0.104 (3)	0.067±0.009 (3)	0.506±0.007 (3)
`mup`	2048	0.100±0.006 (3)	0.153±0.035 (3)	0.065±0.003 (3)	0.506±0.007 (3)
`ctxfree`	2	0.317±0.024 (2)	0.990±0.000 (2)	0.940±0.006 (2)	0.940±0.007 (2)
`ctxfree`	8	0.308±0.023 (3)	0.977±0.005 (3)	0.937±0.002 (3)	0.938±0.002 (3)
`ctxfree`	128	0.304±0.023 (3)	0.937±0.002 (3)	0.941±0.000 (3)	0.941±0.000 (3)
`ctxfree`	512	0.302±0.023 (3)	0.913±0.004 (3)	0.940±0.001 (3)	0.940±0.000 (3)

Runs that did not reach L^*. The following runs ended at the step budget above L^*: `headlr` $M = 32$, $\kappa = 0.000244141$, seed 0 (final loss 0.320 at step 40000); `headlr` $M = 32$, $\kappa = 3.05176e-05$, seed 0 (final loss 0.347 at step 40000).

Table 6: Experiment 2 at the secondary threshold 0.1 (labelled; not pre-registered): mean±sd over seeds that reached it.

arm	M	probe gain	h_u	reversed acc.	context-random acc.
`sp`	2	0.274±0.021 (2)	0.629±0.021 (2)	0.137±0.006 (2)	0.555±0.008 (2)
`sp`	4	0.227±0.006 (3)	0.462±0.055 (3)	0.089±0.009 (3)	0.525±0.007 (3)
`sp`	8	0.241±0.024 (3)	0.514±0.028 (3)	0.099±0.008 (3)	0.531±0.008 (3)
`sp`	32	0.227±0.016 (3)	0.459±0.044 (3)	0.098±0.008 (3)	0.530±0.007 (3)
`sp`	128	0.206±0.011 (3)	0.383±0.044 (3)	0.096±0.010 (3)	0.528±0.009 (3)
`sp`	512	0.170±0.006 (3)	0.282±0.047 (3)	0.096±0.010 (3)	0.528±0.009 (3)
`sp`	2048	0.142±0.007 (3)	0.218±0.038 (3)	0.106±0.013 (3)	0.533±0.011 (3)
`mup`	2	0.281±0.023 (2)	0.669±0.017 (2)	0.138±0.003 (2)	0.555±0.006 (2)
`mup`	4	0.250±0.008 (3)	0.566±0.063 (3)	0.095±0.011 (3)	0.529±0.008 (3)
`mup`	8	0.248±0.003 (2)	0.618±0.051 (2)	0.102±0.014 (2)	0.536±0.009 (2)
`mup`	32	0.243±0.002 (2)	0.591±0.024 (2)	0.089±0.000 (2)	0.528±0.001 (2)
`mup`	128	0.237 (1)	0.579 (1)	0.084 (1)	0.524 (1)
`mup`	512	0.242 (1)	0.611 (1)	0.086 (1)	0.525 (1)

Table 7: Whole-head rate multiplier κ at $M = 32$ (arm `headlr`) at L^*; rows marked † never reached L^* within 40,000 steps and show the final state (loss in parentheses).

seed	κ	step	probe gain	h_u	reversed	context-random	spectral-only
0	3.05176e-05†	40000 (0.347)	0.267	0.701	0.208	0.528	0.689
0	0.000244141†	40000 (0.320)	0.266	0.693	0.184	0.529	0.697
0	0.00390625	22708	0.223	0.475	0.152	0.519	0.641
0	0.0625	7611	0.106	0.177	0.129	0.513	0.568
0	1	1671	0.028	0.079	0.128	0.515	0.536
1	0.00390625	34174	0.203	0.443	0.146	0.510	0.615
1	0.0625	8977	0.100	0.181	0.129	0.509	0.575
1	1	1580	0.025	0.085	0.121	0.511	0.551
2	0.00390625	32068	0.205	0.301	0.130	0.500	0.570
2	0.0625	8704	0.056	0.067	0.110	0.498	0.520
2	1	1881	0.014	0.036	0.112	0.500	0.507

Table 8: Readout learning-rate multiplier at $M = 32$ (arm `lrmult`) at L^*: mean±sd over three seeds.

multiplier	probe gain	h_u	reversed acc.	context-random acc.
0.0625	0.041±0.013	0.085±0.036	0.115±0.009	0.507±0.007
0.25	0.031±0.012	0.076±0.033	0.116±0.009	0.507±0.007
1	0.022±0.007	0.067±0.027	0.121±0.008	0.508±0.008
4	0.015±0.003	0.059±0.022	0.127±0.008	0.509±0.008
16	0.008±0.001	0.054±0.019	0.133±0.009	0.510±0.006

Table 9: Recovery at $M = 32$: a fresh head with paired initialization trained for 20,000 steps on newly sampled context-random images with the encoder frozen at its L^* checkpoint (`init` = the random initial encoder). Original = the run's own classifier at L^*; retrained = the new head; accuracies on the same paired test family.

seed	encoder	probe	retrain loss	orig. reversed	orig. ctx-random	retr. iid	retr. reversed	retr. ctx-random
0	`init`	0.647	0.658	–	–	0.592	0.599	0.596
0	`kappa1`	0.675	0.645	0.128	0.515	0.608	0.618	0.612
0	`kappa256`	0.870	0.328	0.152	0.519	0.845	0.840	0.844
1	`init`	0.658	0.659	–	–	0.596	0.590	0.593
1	`kappa1`	0.683	0.658	0.121	0.511	0.594	0.591	0.592
1	`kappa256`	0.861	0.447	0.146	0.510	0.776	0.771	0.773
2	`init`	0.615	0.682	–	–	0.562	0.533	0.546
2	`kappa1`	0.628	0.676	0.112	0.500	0.574	0.539	0.554
2	`kappa256`	0.820	0.470	0.130	0.500	0.757	0.750	0.755

Table 10: Registered predictions (REFOCUS_PLAN §4.4) evaluated at $L^* = 0.30$ on the original five widths and on the amended grid; ρ = Spearman rank correlation across widths (or multipliers), one value per seed. P2's second clause compares a_u at $M = 2048$.

criterion	grid	per-seed values	verdict
P1: standard param., $\rho(a_u, M) \leq -0.8$ and $\rho(\text{rev}, M) \leq -0.8$ in every seed	original	a_u: -0.70, -1.00, -1.00; rev: -0.60, -0.90, -0.60	not met (1/3 seeds)
	amended	a_u: -0.54, -1.00, -0.96; rev: -0.64, -0.96, -0.82	not met
P2a: normalized readout, $\lvert\rho(\cdot, M)\rvert < 0.5$ in at least two seeds	original	a_u: +0.10, -0.90, -0.30; probe: -0.20, -0.90, -0.30; rev: -0.90, -0.70, -0.70	alignment met; probe met; reversal not met
	amended	a_u: +0.25, -0.96, -0.54; rev: -0.89, -0.89, -0.89	alignment not met; reversal not met
P2b: $a_u(2048)$ normalized $>$ standard in every seed	paired seeds	0.0059 vs 0.0045; 0.0062 vs 0.0045; 0.0043 vs 0.0027	met
P3: uninformative context, $a_u > 0.25$ at every width and $\lvert\rho(a_u, M)\rvert < 0.5$	amended control widths $\{2, 8, 128, 512\}$	a_u range 0.058–0.174; $\rho(a_u, M)$: -1.00, -1.00, -1.00; $\rho(\text{probe}, M)$: -1.00, -1.00, -1.00	cutoff not met; flatness not met (probe 0.936→0.893)
P4: readout multiplier, a_u and reversal accuracy decrease with the multiplier in every seed	run multipliers $\{1/16, 1/4, 1, 4, 16\}$ (64 dropped)	a_u: -1.00, -1.00, -1.00; rev: +1.00, +1.00, +1.00	a_u met; reversal not met (opposite sign)
P5: frozen encoder, reversal accuracy < 0.5 at every width	amended widths	max 0.146	met
P6 (amendment 4.4b): whole-head κ, probe, h_u and reversal accuracy increase as κ decreases, $\rho \leq -0.8$ in every seed	$\kappa \in \{1, 1/16, 1/256\}$, three seeds	probe: -1.00, -1.00, -1.00; h_u: -1.00, -1.00, -1.00; rev: -1.00, -1.00, -0.50	probe and h_u met; reversal not met (2/3 seeds)

D Experiment 3 details

D.1 Data

Breast-tissue QCL micro-array cores, fold 0 of the five-fold patient-level split used throughout this work (115 training, 28 validation, 26 test cores). Each labelled tissue pixel contributes the 942-dimensional vector of absorbance, first and second derivative (3×314 channels). Up to 600 (training) or 300 (validation, test) pixels per class per core were sampled with a fixed seed: 75,297 training pixels from 79 cores, 11,024 validation from 20, 8,038 test from 16, over the four classes; the experiment uses the pair Cancer Epithelium ($+1$; 29,603 training pixels) versus Cancer-Associated Stroma (-1; 21,923).

D.2 Readiness scan

For each class pair, bottleneck K and preprocessing, Table 11 reports the centre-only oracle (ridge Fisher discriminant on the preprocessed 942-vector, fitted on one half of the training cores and evaluated on the other half) and the same discriminant through three random encoders $W \sim \mathcal{N}(0, 1/942)^{K \times 942}$. Standardization is per feature; whitening is PCA whitening with a relative ridge of 10^{-3} on the eigenvalues; both are estimated on the fitting half.

Table 11: Readiness scan (`results/exp3/readiness_scan.csv`): centre-only oracle and random-encoder probe (mean over three encoders) by pair, K and preprocessing. $\lvert\cos\rvert$ is the cosine between the class-mean contrast and the first principal direction of the standardized training half.

pair	preprocessing	oracle	$K = 2$	$K = 4$	$K = 12$	$K = 32$	$\lvert\cos\rvert$
CancerEpi vs CAS	standardized	0.969	0.726	0.803	0.955	0.968	0.50
CancerEpi vs CAS	whitened	0.944	0.503	0.486	0.539	0.591	0.50
NormalStroma vs CAS	standardized	0.810	0.580	0.639	0.674	0.741	0.44
NormalStroma vs CAS	whitened	0.768	0.502	0.509	0.529	0.600	0.44
NormalEpi vs CancerEpi	standardized	0.758	0.547	0.573	0.662	0.724	0.20
NormalEpi vs CancerEpi	whitened	0.761	0.575	0.545	0.624	0.580	0.20
NormalStroma vs CancerEpi	standardized	0.989	0.781	0.874	0.978	0.990	0.88
NormalStroma vs CancerEpi	whitened	0.974	0.579	0.541	0.549	0.626	0.88

With standardized inputs a random encoder already reads the epithelium-versus-stroma contrasts, because they lie along the top principal directions; whitening removes that accessibility while the oracle stays high. The two regimes of Section 5 follow: the same pair, $K = 12$, standardized (*ready*) or whitened (*unready*).

D.3 PATCH CONSTRUCTION

Each training example is a real centre pixel with its recorded label and eight real donor spectra sampled uniformly from the same split side (resampled once if the donor's core equals the centre's), independent of the centre label. In the preprocessed space, donor d receives the constructed cue $\gamma_d\,(c\,s + \tau\eta_d)\,e_d$, where $e_1, \ldots, e_8$ are the first eight principal coordinates (in the standardized regime the directions $v_1, \ldots, v_8$ with $\gamma_d = 3\sqrt{\lambda_d}$; in the whitened regime the first eight whitened coordinates with $\gamma_d = 30$), c is the centre's label code, and $s = +1$ (informative), $s = -1$ (reversed) or c replaced by an independent random label (context-random); *spectral-only* removes the cue and keeps the donors. The draw order is condition-independent, so the four conditions generated from one seed share centres, donors and η. Preprocessing, principal directions and λ_d are estimated on training pixels only. Training: 24,000 class-balanced centres; probe fitting: 6,000 disjoint training pixels; evaluation: up to 6,000 validation and 6,000 test centres with donors from their own side. τ is calibrated per regime on training-core halves so that the neighbourhood oracle (a discriminant on the eight cue projections) matches the centre-only oracle.

Table 12: Experiment 3 calibration (`results/exp3/calibration3.json`). Readiness: discriminant through a random 12×942 encoder on the centre alone and on the encoded 3×3 patch (three encoders).

regime	τ	centre oracle (fit/val)	context oracle	random enc. centre	random enc. patch
ready (standardized)	1.475	0.969 / 0.959	0.969	0.935, 0.970, 0.952	0.988, 0.994, 0.991
unready (whitened)	1.735	0.948 / 0.884	0.948	0.476, 0.512, 0.605	0.948, 0.947, 0.950

D.4 MODEL, TRAINING AND ARMS

Encoder $W \in \mathbb{R}^{12\times 942}$ applied to the nine spectra; head $h = \mathrm{ReLU}(W_1 \operatorname{vec}(z) + b_1)$ of width M over the $9K = 108$ encoded coordinates and a linear readout to one logit; margin loss; full-batch gradient descent with $\eta = 10^{-3}$, at most 40,000 steps, the same thresholds and snapshot rule as Experiment 2. Arms per regime: `sp` at $M \in \{8, 32, 128, 512\}$; `headlr` with $\kappa \in \{1, 1/16, 1/256\}$ at $M = 32$; `ctxfree` (context-random training) at $M \in \{8, 128\}$; `frozen` at $M = 32$; seeds 0–2. Spectral probe: ridge discriminant on the encoder output of centre pixels, fitted on the probe set and evaluated on validation (and test) centres, reported as the gain over its initial value. Reliance and failure are measured on validation patients with the paired conditions; test patients are reported separately and were not used for any choice.

D.5 RESULTS

Tables generated by `paper/figures_src/make_tables_exp3.py` from `results/exp3_summary.csv`.

Runs that did not reach L^*. Every run reached L^*.

E SCOPE LIMITATIONS FROM A PRODUCTION MODEL

This appendix reproduces, from the earlier diagnostic draft of this work, the measurements on a production spectral–spatial segmentation model that are summarized in Section 6. They concern initialization curvature and a controlled retraining study; they do not measure the mechanism of Theorem 1 and are included to document why scalar curvature is not the diagnostic used in this paper and why the earlier apparent freezing advantage is not evidence for or against it. The data are breast-tissue QCL micro-array cores with a prostate QCL replication. Theorem and proposition numbers quoted inside this appendix refer to the earlier draft's framework and are retained only where the sentence is a scope statement about that framework.

Table 13: Experiment 3, unready (whitened, $\gamma = 30$) regime, at $L^* = 0.30$: mean±sd over seeds (number of seeds). Probe gain = discriminant accuracy on the encoder output of validation centres minus its initial value; accuracies on validation (val) and test (test) patients under the paired conditions.

arm	M / κ	step	probe gain	rev. (val)	ctx-rand. (val)	spec.-only (val)	rev. (test)	ctx-rand. (test)
`sp`	8	126±51	0.015±0.031	0.083±0.014	0.500±0.012	0.485±0.028	0.083±0.005	0.496±0.000 (3)
`sp`	32	113±22	0.032±0.042	0.072±0.006	0.500±0.010	0.510±0.006	0.070±0.003	0.495±0.001 (3)
`sp`	128	100±12	0.026±0.017	0.075±0.003	0.500±0.010	0.495±0.015	0.073±0.006	0.496±0.004 (3)
`sp`	512	69±8	-0.012±0.022	0.066±0.001	0.498±0.011	0.486±0.010	0.067±0.005	0.495±0.004 (3)
`headlr`	32 / 0.00390625	185±43	0.046±0.052	0.075±0.012	0.500±0.013	0.512±0.011	0.072±0.002	0.494±0.001 (3)
`headlr`	32 / 0.0625	177±40	0.044±0.051	0.074±0.011	0.500±0.013	0.512±0.010	0.073±0.001	0.495±0.002 (3)
`headlr`	32	113±22	0.032±0.042	0.072±0.006	0.500±0.010	0.510±0.006	0.070±0.003	0.495±0.001 (3)
`ctxfree`	8	16780±2138	0.352±0.008	0.925±0.015	0.923±0.015	0.924±0.010	0.935±0.023	0.934±0.023 (3)
`ctxfree`	128	14954±488	0.336±0.014	0.911±0.030	0.908±0.029	0.926±0.028	0.930±0.014	0.928±0.009 (3)
`frozen`	32	395±95	0.000±0.000	0.076±0.006	0.500±0.009	0.515±0.008	0.076±0.002	0.496±0.002 (3)

Table 14: Experiment 3, unready (whitened, $\gamma = 10$; amendment) regime, at $L^* = 0.30$: mean±sd over seeds (number of seeds). Probe gain = discriminant accuracy on the encoder output of validation centres minus its initial value; accuracies on validation (val) and test (test) patients under the paired conditions.

arm	M / κ	step	probe gain	rev. (val)	ctx-rand. (val)	spec.-only (val)	rev. (test)	ctx-rand. (test)
`sp`	32	592±70	0.076±0.055	0.067±0.004	0.501±0.011	0.512±0.007	0.063±0.002	0.496±0.002 (3)
`headlr`	32 / 0.00390625	1579±318	0.134±0.072	0.066±0.010	0.501±0.011	0.508±0.015	0.065±0.005	0.496±0.001 (3)
`headlr`	32 / 0.0625	1337±227	0.124±0.068	0.065±0.009	0.501±0.011	0.508±0.013	0.065±0.003	0.496±0.001 (3)
`headlr`	32	592±70	0.076±0.055	0.067±0.004	0.501±0.011	0.512±0.007	0.063±0.002	0.496±0.002 (3)
`ctxfree`	32	16003±231	0.350±0.012	0.916±0.005	0.916±0.006	0.925±0.009	0.923±0.011	0.924±0.011 (3)
`frozen`	32	2006±310	0.000±0.000	0.080±0.008	0.502±0.011	0.511±0.007	0.073±0.004	0.495±0.001 (3)

Table 15: Experiment 3, ready (standardized) regime, at $L^* = 0.30$: mean±sd over seeds (number of seeds). Probe gain = discriminant accuracy on the encoder output of validation centres minus its initial value; accuracies on validation (val) and test (test) patients under the paired conditions.

arm	M / κ	step	probe gain	rev. (val)	ctx-rand. (val)	spec.-only (val)	rev. (test)	ctx-rand. (test)
`sp`	8	132±59	0.000±0.017	0.087±0.004	0.502±0.017	0.522±0.035	0.088±0.009	0.499±0.003 (3)
`sp`	32	118±22	0.002±0.013	0.083±0.013	0.502±0.013	0.526±0.026	0.088±0.013	0.500±0.003 (3)
`sp`	128	103±20	0.006±0.016	0.102±0.004	0.512±0.011	0.578±0.060	0.102±0.009	0.511±0.008 (3)
`sp`	512	70±3	0.002±0.014	0.092±0.012	0.508±0.011	0.544±0.017	0.092±0.021	0.504±0.005 (3)
`headlr`	32 / 0.00390625	193±31	0.003±0.013	0.083±0.010	0.503±0.013	0.525±0.028	0.086±0.008	0.501±0.001 (3)
`headlr`	32 / 0.0625	184±29	0.002±0.013	0.084±0.011	0.503±0.013	0.524±0.028	0.086±0.008	0.501±0.001 (3)
`headlr`	32	118±22	0.002±0.013	0.083±0.013	0.502±0.013	0.526±0.026	0.088±0.013	0.500±0.003 (3)
`ctxfree`	8	1128±116	0.015±0.021	0.929±0.002	0.923±0.001	0.931±0.005	0.953±0.008	0.947±0.004 (3)
`ctxfree`	128	841±112	0.012±0.021	0.927±0.006	0.925±0.006	0.929±0.004	0.955±0.001	0.954±0.000 (3)
`frozen`	32	442±114	0.000±0.000	0.091±0.022	0.504±0.013	0.529±0.021	0.093±0.027	0.499±0.006 (3)

E.1 Production model details

E.1.1 Pipeline, capacities and data

The data are 336×336 pixel tissue-microarray cores collected by quantum-cascade-laser infrared spectroscopy over the 1000–1800 cm^{-1} fingerprint region, with per-pixel four-class tissue labels. Each pixel carries an $S = 942$ dimensional vector, obtained by stacking baseline-corrected absorbance with its first two derivatives. The compositional model is the Block Vision Transformer of O'Leary et al. (2026): a linear per-pixel $f_{\boldsymbol{\theta}} : \mathbb{R}^{942} \to \mathbb{R}^K$, then 16×16 patch tokenization, a 12-layer, 12-head vision transformer of embedding width M, a transposed-convolution decoder and a convolutional segmentation head. Parameter counts in Table 16 are read from the instantiated model.

Table 16: Module capacities of the production pipeline, counted from the instantiated model. K is the spectral bottleneck width. The last row is the single-channel field-standard input convention, for comparison.

Configuration	C_f	C_g	C_g/C_f
$K = 64$, three-channel input	61,108	18,271,540	299
$K = 128$, three-channel input	121,588	21,472,564	177
$K = 64$, single-channel input	20,916	18,271,540	874

Channel convention. The published architecture operates on the second-derivative spectrum alone, the field-standard preprocessing in infrared histopathology. The models analysed here instead stack raw absorbance with its first two derivatives, following a companion tokenization study (anonymous, under review). The choice affects only C_f, and the three-channel convention is the conservative one for the present purpose: under the field standard the capacity asymmetry is roughly three times larger. Within the spatial module the dominant blocks are the transformer (5.34×10^6 parameters) and the transposed convolution (9.44×10^6), both $\Theta(M^2)$ in the embedding width.

The head hypothesis is structurally unmet. The segmentation head is the chain $\mathrm{Conv}_{3\times3}(M{+}K \to 96) \to \mathrm{BN} \to \mathrm{ReLU} \to \mathrm{Conv}_{3\times3}(96 \to 48) \to \mathrm{BN} \to \mathrm{ReLU} \to \mathrm{Conv}_{3\times3}(48 \to 4)$, so the final affine classifier sees $48 \times 3 \times 3 = 432$ input features at every width we build ($M \in \{48, 192, 384\}$, $K \in \{64, 128\}$): the $\Theta(M)$ feature map is separated from the logits by two nonlinearities. The curvature-disparity theorem requires a dense classifier over penultimate features of dimension $\Theta(M)$ with no downstream nonlinearity, and that hypothesis is not satisfied here. Failing a hypothesis removes a guarantee; it does not reverse the conclusion.

E.1.2 Geometry at initialization

Gauss–Newton blocks of the instantiated pipeline were measured on real training batches at random initialization, by Lanczos with full reorthogonalization on matrix-free GGN block operators, validated to 10^{-12} against dense computation, sweeping $M \in \{48, 96, 192, 384\}$ with three seeds and four batches per width. Table 17 collects the results.

Two controls bracket the denominator. Removing the spectral normalization deflates $\lambda_{\max}(G_{\boldsymbol{\theta\theta}})$ by 2.4× ($471 \to 194$ at $M = 192$) and pushes the ratio across parity (1.18 at $M = 192$, 1.31 at $M = 384$). Input *centering* changes nothing when the architecture contains internal batch normalization: $\mathcal{D}_{\mathrm{curv}}$ is identical to three decimals with and without centering, because a linear projection followed by normalization absorbs constant input shifts exactly. The near-rank-one input statistics make the cap side of the curvature-disparity bound uninformative at buildable widths independently of the head-hypothesis failure: the data concentrate almost all their second-moment mass in one direction, and that direction is the mean spectrum. The directional measurements establish an anisotropy, not its consequence for learning; whether it starves learning is a hypothesis, not a measurement.

E.1.3 What carries the width trend

The measurements in this subsection are made on the production model at random initialization, at three widths $M \in \{48, 192, 384\}$ and three seeds, on one core (the densest of the

Table 17: Production geometry at initialization. Ratios are mean ± sd over three seeds and four batches per width. Slopes are log–log fits over the width sweep. Effective ranks are out of $S = 942$. Directional entries are measured at $M = 192$ over the nine measurements whose batch contains both classes (three seeds × three batches).

Quantity	Breast	Prostate
$\mathcal{D}_{\text{curv}}$ at $M = 48$	0.33 ± 0.12	—
$\mathcal{D}_{\text{curv}}$ at $M = 192$	0.46 ± 0.21	—
$\mathcal{D}_{\text{curv}}$ at $M = 384$	0.70 ± 0.19	1.09 ± 0.52
log–log slope of $\lambda_{\max}(G_{\boldsymbol{\theta\theta}})$ vs M	-0.003	-0.012
log–log slope of $\lambda_{\max}(G_{\boldsymbol{\phi\phi}})$ vs M	0.38 ($115 \to 258$)	0.45
Effective rank of Σ_X, raw inputs	1.07	1.02
Effective rank of Σ_X, post-normalization	1.21	—
$\lvert\cos\rvert$ between v_1 and the mean spectrum (post-norm. / raw)	> 0.995 / > 0.9999	—
$\boldsymbol{\theta}$-block top eigenvector's norm fraction in the v_1-induced subspace	0.954 ± 0.017	—
Trace fraction in the top 40 of 61,108 spectral directions	0.83 ± 0.12	—
Curvature ratio, v_1 vs full class contrast ($\lvert\cos(d, v_1)\rvert = 0.45$)	3.7× (2.6–4.8)	—
Curvature ratio, v_1 vs v_1-orthogonalized contrast	12–28×	—
Squared-norm fraction of the contrast retained after orthogonalization	76–84%	—
Rank at which $\lambda_{\max}(G_{\boldsymbol{\phi\phi}})$ overtakes the $\boldsymbol{\theta}$ spectrum	2–5	—

first 24 breast fold-0 training cores, with 29,664 labelled pixels of 112,896), with cuDNN TF32 disabled. Four normalization modes are compared: `train` (both head batch normalizations using batch statistics), `eval_bn1` and `eval_bn2` (only the first or only the second head normalization at its initial running statistics) and `eval_head` (both). Artifacts: `results/exp1_8c_interior_witness.csv` and `results/exp1_8d_REPORT.md`.

The incoming features are flat in width. The only downstream layer whose input dimension grows with M is the first head convolution. The top eigenvalue of the patch-unfolded second-moment matrix of the features entering it is unchanged to four significant figures across the three widths, at 356.3, 403.7 and 332.4 for seeds 0, 1 and 2 respectively; the leading direction sits inside the width-independent $K = 64$ skip channels. Table 18 also records a discrepancy that matters for any normalization argument: the same Gram computed over the full batch-normalization group (all positions, ignored pixels included) rather than over labelled pixels is substantially smaller, and the cosine between the two centered top eigenvectors is far from one.

Table 18: Incoming 3×3-patch Gram of the first head convolution, top eigenvalue, from `results/exp1_8d_REPORT.md`. Values are identical across $M \in \{48, 192, 384\}$ and across all four normalization modes, so a single column suffices. *labelled* uses the labelled-pixel mask; *full group* uses the batch-normalization group. The last column is the cosine between the two centered top eigenvectors.

seed	labelled	labelled, centered	full group	cos(labelled, full)
0	356.3	68.19	263.3	0.2793
1	403.7	54.60	306.8	0.5502
2	332.4	67.14	253.0	0.2176

Normalization quantities. Table 19 gives the per-channel batch statistics of the two head normalizations, measured over the group they actually normalize, together with the activation energy entering and leaving the first one. Under fan-in initialization the pre-normalization variance of the first head normalization falls roughly as $1/(M + K)$, and its inverse-standard-deviation gain grows correspondingly. No channel's variance is within a factor of ten of the stabilizer $\varepsilon = 10^{-5}$: the fraction of channels with variance below 10ε is 0 in every row of both normalizations at every width, seed and mode. (This recorded check does not establish the stronger separation by three orders of magnitude claimed in the earlier draft.)

Table 19: Head normalization quantities at initialization, `train` mode, from `results/exp1_8d_REPORT.md` Table 1. Per-seed values are per-channel medians (variance, gain) or labelled-pixel energies; the *mean* column is the mean of the three seed values, computed by us. $\mathrm{gain}_i = \gamma_i/\sqrt{\mathrm{var}_i + \varepsilon}$. The last row is the product of the mean variance with $M + K$, $K = 64$.

Quantity	seed 0	seed 1	seed 2	mean	M
First head BN, median variance	0.1470	0.1467	0.1587	0.1508	48
	0.0678	0.0589	0.0656	0.0641	192
	0.0346	0.0310	0.0398	0.0351	384
First head BN, median gain	2.609	2.611	2.511	2.577	48
	3.842	4.120	3.905	3.956	192
	5.375	5.682	5.010	5.356	384
Second head BN, median gain	2.892	3.122	3.039	3.018	48
	3.110	3.177	3.195	3.161	192
	2.966	3.060	3.121	3.049	384
Pre-BN energy, labelled pixels	25.64	26.28	23.91	25.28	48
	11.07	10.67	11.19	10.98	192
	5.765	5.543	6.015	5.774	384
Post-BN energy, labelled pixels	132.1	130.8	133.2	132.0	48
	130.4	131.5	132.5	131.5	192
	130.7	128.5	130.8	130.0	384
Mean variance × $(M{+}K)$	—	—	—	16.9 / 16.4 / 15.7	48 / 192 / 384

Witness Jacobian-vector products. A unit perturbation of the first head convolution is propagated by forward-mode differentiation through the actual head in the actual normalization mode, so that the train-mode derivative includes the dependence on the batch statistics. The final scalar $e_{\mathrm{final}} = N^{-1}\sum_n d_n^\top(\mathrm{diag}(p_n) - p_n p_n^\top)d_n$ is a lower-bound witness for $\lambda_{\max}(G_{\boldsymbol{\phi\phi}})$ and is verified against the same matvec the eigensolver uses; all 216 rows pass at relative discrepancy below 10^{-3}, with a maximum of 1.269×10^{-5} over the 198 rows with nonzero curvature. Table 20 gives the stage energies for two directions in two modes.

Table 20: Witness JVP stage energies, mean over three seeds, from `results/exp1_8d_REPORT.md` Table 2a. `ggn_top` is the full top GGN eigenvector of the convolution block; `s9cen` is the centered leading direction of the incoming patch Gram. e_{pre} is the perturbation energy before the first head normalization, e_{logit} is measured on labelled pixels.

Mode	Direction	M	e_{pre}	e_{bn1}	e_{bn2}	e_{logit}	e_{final}
`train`	`ggn_top`	48	260.6	2875	4453	261.7	60.73
`train`	`ggn_top`	192	259.4	5905	7553	505.2	118.5
`train`	`ggn_top`	384	256.8	9168	13810	1018	227.6
`train`	`s9cen`	48	76.53	1538	1224	29.23	5.761
`train`	`s9cen`	192	76.53	2305	1914	50.83	9.862
`train`	`s9cen`	384	76.53	3997	3906	124.3	21.95
`eval_head`	`ggn_top`	48	261.9	261.9	42.42	9.607	2.366
`eval_head`	`ggn_top`	192	262.8	262.8	42.18	10.35	2.487
`eval_head`	`ggn_top`	384	262.5	262.5	46.53	10.45	2.372
`eval_head`	`s9cen`	48	76.53	76.53	9.103	0.4667	0.1049
`eval_head`	`s9cen`	192	76.53	76.53	8.460	0.4613	0.1013
`eval_head`	`s9cen`	384	76.53	76.53	9.762	0.4450	0.1011

A spatially constant bias perturbation $\Delta b = e_c$ is annihilated exactly by the immediately following train-mode normalization: in `train` and `eval_bn2` modes, at every width and seed, all stage energies after the centering step are exactly zero and the quadratic form agrees with the matvec to at most 1.14×10^{-17} in absolute value. In `eval_head` the same perturbation gives e_{final} between 7.8×10^{-4} and 2.0×10^{-3}, and in `eval_bn1` between 0.0154 and 0.0692.

Slopes and attribution. Table 21 gives the log–log slopes against M per seed and pooled, together with the mode-to-mode ratios of levels.

Table 21: Log–log slopes against M over the three widths, per seed and pooled, and mode-to-mode ratios of the levels (each mode's value divided by its train-mode value at the same width and seed, then averaged over seeds and over the three widths), from `results/exp1_8d_REPORT.md`. λ_{WW} is the top curvature of the first head convolution's weight block; λ_{ϕ} is the whole spatial block. Three-width slopes are descriptive, not asymptotic exponents.

	slope of λ_{WW}				λ_{ϕ}	level ratio vs `train`	
Mode	seed 0	seed 1	seed 2	pooled	pooled slope	λ_{WW}	λ_{ϕ}
`train`	0.509	0.785	0.350	$+0.613$	$+0.398$	1	1
`eval_bn1`	0.429	0.634	0.542	$+0.563$	$+0.281$	0.903	1.03
`eval_bn2`	0.544	0.632	0.508	$+0.571$	$+0.253$	0.156	0.227
`eval_head`	-0.112	0.181	-0.084	$+0.006$	-0.331	0.026	0.037

Three readings, and one non-reading. The positive width trend in the spatial block's curvature at initialization is sensitive to head batch normalization: holding both head normalizations at their initial running statistics removes it ($+0.613 \to +0.006$ for the convolution block, $+0.398 \to -0.331$ for the whole spatial block) while leaving the incoming features unchanged. Either normalization alone preserves the trend ($+0.563$ and $+0.571$), which is evidence against attributing it uniquely to the first one, even though most of the drop in the *level* attaches to the second. The mechanism this supports is a finite-range normalization gain acting on width-diluted activations, over the range of widths measured, and not the width-linear feature-Gram mechanism of the shallow theory. The non-reading: the `eval_head` intervention changes the normalization derivative, the forward logits, the downstream gates and the cross-entropy metric at the same time, so a difference between modes establishes sensitivity to the intervention, not a unique cause; and the observation that the variance never approaches the stabilizer is a heuristic about the regime, not a guarantee.

Parameterization control. Table 22 scales the weights and bias of the first head convolution by c and, in the exact invariance, the following stabilizer by c^2. The logits are unchanged and the block's top curvature scales by exactly c^{-2}: the raw Euclidean curvature of a layer feeding a normalization is a parameterization quantity, not an architectural one (van Laarhoven, 2017).

Table 22: Function-preserving scale control at seed 0, from `results/exp1_8d_REPORT.md` Table 3. *ratio* is $c^2\lambda_{WW}(c)/\lambda_{WW}(1)$, which is 1 under exact invariance. *logit dev.* is $\max|\text{logits}(c) - \text{logits}(1)|$, reported both with the stabilizer co-scaled by c^2 and with it held fixed (the implementation's actual behaviour); λ_{WW}, *ratio* and $\|W\|_F^2$ are taken from the fixed-stabilizer variant, and the co-scaled variant differs from it by at most one unit in the fourth significant figure.

M	c	λ_{WW}	ratio	logit dev. (co-scaled)	logit dev. (fixed ε)	$\|W\|_F^2$
48	1/4	719.4	1.000	0	$9.40 \cdot 10^{-4}$	2.003
48	1/2	179.8	1.000	0	$1.89 \cdot 10^{-4}$	8.014
48	1	44.96	1.000	0	0	32.06
48	2	11.24	1.000	0	$4.63 \cdot 10^{-5}$	128.2
48	4	2.810	1.000	0	$5.84 \cdot 10^{-5}$	512.9
192	1/4	1935	0.9998	0	$3.00 \cdot 10^{-3}$	2.002
192	4	7.560	1.000	0	$1.89 \cdot 10^{-4}$	512.6
384	1/4	1927	0.9991	0	$4.69 \cdot 10^{-3}$	1.999
384	4	7.535	1.000	0	$2.95 \cdot 10^{-4}$	511.8

Eigensolver accuracy. Every row of the width sweep terminates on the combined criterion (relative change of the Rayleigh quotient below tolerance *and* residual below 10^{-3}), never on the iteration or wall-clock budget. Over all 36 (mode, width, seed) rows the relative residual $\|Gv - \lambda v\| / \lambda$ is at most 3.81×10^{-4} for the convolution weight block, at most 3.81×10^{-4} for the weight–bias block and at most 9.97×10^{-4} for the whole spatial block, and the Rayleigh quotient of the returned iterate agrees with the reported eigenvalue to the printed precision in every row.

E.1.4 Controlled retraining: original protocol

The pipeline was retrained on breast fold 0 at reduced depth (six transformer layers) with full instrumentation. Two protocols were run and both are reported: the *original* protocol (39 runs), which is the pre-registered one and whose joint-versus-frozen contrast an audit of the training code later showed to be confounded, and the *matched* protocol (21 runs), which removes the confounds, saves the selected checkpoint and re-runs the pre-registered comparisons. Sixty logged runs in total. Per-step residuals, block gradient norms, per-epoch parameter displacements and input-Jacobian norms are recorded in both; the matched protocol additionally logs the per-step clipping coefficient and the per-block applied update norms.

The original protocol comprises arms {joint-linear, frozen-random, frozen-PCA, joint-MLP} $\times$ widths $M \in \{48, 192\}$ $\times$ three seeds under AdamW for 60 epochs, plus a {joint-linear, frozen-random} $\times$ three-seed pair under momentum SGD at both widths and a frozen-PCA SGD triple at $M = 48$ as a positive control. The spectral parameters sit in a zero-weight-decay group in every arm, and frozen arms log counterfactual spectral gradients. Table 23 gives the AdamW arms.

Table 23: Controlled study, original protocol, AdamW arms: validation macro-F1 (mean $\pm$ sd over three seeds) under two checkpoint policies, and relative spectral displacement at end of training. *Final-5* is the pre-registered metric. *Best-val* is the maximum over the 60 per-epoch evaluations on the same 28 validation cores; no checkpoint was saved at that maximum and this study has no test set, so that column is exploratory and selection-optimistic. The two policies order the arms oppositely. These arms also differ in clip scope and in normalization-affine treatment, so this contrast is confounded.

	Final-5 (pre-registered)		Best-val (exploratory)		
Arm	M=48	M=192	M=48	M=192	$\|\Delta\boldsymbol{\theta}\| / \|\boldsymbol{\theta}_0\|$
Joint (linear)	0.538 ± 0.082	0.646 ± 0.050	0.873 ± 0.030	0.858 ± 0.056	0.54–0.64
Frozen random	0.648 ± 0.063	0.731 ± 0.023	0.840 ± 0.025	0.817 ± 0.017	0
Frozen PCA	0.731 ± 0.003	0.726 ± 0.012	0.769 ± 0.021	0.862 ± 0.068	0
Joint (MLP)	0.566 ± 0.070	0.575 ± 0.184	0.900 ± 0.000	0.897 ± 0.001	0.85–0.96

On the pre-registered final-window metric the joint-linear minus frozen-random gaps are -0.110 at $M = 48$ (90% paired-t interval $[-0.333, +0.114]$) and -0.085 at $M = 192$ ($[-0.185, +0.015]$), neither resolvable at $n = 3$; against frozen-PCA they are -0.193 ($[-0.334, -0.052]$) and -0.081 ($[-0.175, +0.014]$). Taking each run's maximum over its 60 evaluations reverses the ordering: joint minus frozen-random becomes $+0.033$ and $+0.041$, joint minus frozen-PCA $+0.104$ and -0.004. The switch is an arithmetic identity — the change in every gap equals the difference in peak-to-final degradation between the two arms — and averaged over seeds that degradation is 0.335 ($M = 48$) and 0.212 ($M = 192$) for joint-linear against 0.192 and 0.086 for frozen-random.

Three asymmetries. First, gradient-norm clipping at 1.0 was computed over the optimizer's parameters, that is over $\boldsymbol{\theta} + \boldsymbol{\phi}$ in the joint arms but over $\boldsymbol{\phi}$ alone in the frozen arms, so at identical gradients the frozen arm takes the larger $\boldsymbol{\phi}$ step: the median extra clip factor paid by the joint arm is 1.10–1.12 under SGD at $M = 48$ and 3.0–3.6 under AdamW at $M = 192$. Second, the normalization following the spectral projection contributes 128 affine parameters that are trained in joint-linear, frozen in frozen-random and structurally absent in frozen-PCA. Third, no checkpoint was saved during training, so the best-validation column is a post-hoc curve maximum that no saved model realizes, taken over 60 evaluations of the same 28 validation cores that define the comparison.

E.1.5 Momentum-SGD controls and normalization recalibration

The two optimizer families constitute an optimizer-family control; the SGD arm is not an instantiation of the two-timescale theorem's gradient flow. It uses momentum 0.9; weight decay 0.01 on $\boldsymbol{\phi}$ and 0 on $\boldsymbol{\theta}$; mini-batches of four cores with reshuffling; and global gradient-norm clipping at 1.0, which binds on 73–89% of steps. Logged gradient norms are recorded before clipping, so the plotted theory quantities are not the quantities the trajectory applied. Under momentum SGD the paired joint minus frozen-random gap is $+0.002$ at $M = 48$ and $+0.011$ at $M = 192$ on best-validation, with 90% paired-t intervals $[-0.008, +0.011]$ and $[-0.007, +0.030]$; on the final-window metric the same pairs give $+0.028$ and -0.057, with intervals $[-0.173, +0.228]$ and $[-0.124, +0.010]$ — gaps that differ in sign between the widths, with intervals an order of magnitude wider than the effect they bracket. Both arms lose 0.12–0.22 macro-F1 between their peak and their final window. Descriptively, the spectral share of the summed squared gradient norms drops from about 0.94 (AdamW joint) to about 0.26 (SGD joint); momentum, clipping, weight decay and adaptivity all differ between the two families, so no single factor is isolated.

Recalibration of normalization statistics. On disposable copies of the final checkpoints, with all weights fixed, the normalization running statistics were re-estimated on training data only (29 batches) and the model re-evaluated. For this intervention and these checkpoints, the recovered fraction of the joint arms' peak-to-final drop is 0.17 at $M = 48$ and 0.07 at $M = 192$: recalibration recovers a nonzero part of the degradation but not most of it. What it does undo is the instability of the endpoint number: it repairs the two collapsed joint checkpoints ($+0.24$ and $+0.15$ macro-F1) and shrinks the seed standard deviation by roughly $3.7\times$, while leaving the frozen arms essentially unchanged ($|\Delta| \leq 0.02$, and ≤ 0.006 for frozen-PCA, whose spectral stage carries no normalization).

E.1.6 Matched protocol

The matched protocol equalizes the three asymmetries and re-runs the pre-registered comparisons at $M = 48$: the reduction-stage normalization affine parameters are placed in $\boldsymbol{\phi}$ and trained wherever that normalization exists; the clip norm at 1.0 is computed over $\boldsymbol{\phi}$ only under the same clipping rule and scope in every arm; and the best-validation checkpoint is saved. AdamW at learning rate 10^{-4}, 60 epochs, three seeds, flip and $90°$-rotation augmentation on in all arms. The logged per-step clip coefficient averages 0.33–0.57 across arms — the coefficients and gradients differ by arm, which is an outcome of the matched rule — and the logged applied update norms confirm that the spectral share of the applied update tracks the learning-rate multiplier (0.02, 0.17 and 1.36 times the $\boldsymbol{\phi}$ update norm for $\times 0.1$, $\times 1$ and $\times 10$). Table 24 gives all seven arms.

Table 24: Controlled study, matched protocol ($M = 48$, AdamW, three seeds, mean $\pm$ sd). Identical normalization-affine treatment and the same $\boldsymbol{\phi}$-only clipping rule and scope in every arm; best-validation checkpoints saved. *Final-5* is the pre-registered metric; *best-val* is the maximum over 60 per-epoch evaluations on the 28 fold-0 validation cores and is exploratory and selection-optimistic, since this study has no test set. The drift column is the relative spectral-parameter displacement at end of training; its standard deviations were computed by us from `results/e3c_analysis_runs.csv`. Rows 1–2 and rows 3–4 are the two matched pairs; see the text.

Arm	Final-5 (pre-registered)	Best-val (exploratory)	$\lVert\Delta\boldsymbol{\theta}\rVert \,/\, \lVert\boldsymbol{\theta}_0\rVert$
Frozen random	0.684 ± 0.037	0.814 ± 0.040	0
Joint (linear)	0.694 ± 0.056	0.837 ± 0.029	0.566 ± 0.026
Frozen pretrained	0.690 ± 0.033	0.861 ± 0.022	0
Fine-tuned pretrained	0.710 ± 0.052	0.857 ± 0.009	0.020 ± 0.005
Joint, spectral lr $\times 0.1$	0.626 ± 0.055	0.841 ± 0.007	0.083 ± 0.006
Joint, spectral lr $\times 10$	0.561 ± 0.114	0.886 ± 0.037	4.910 ± 0.402
Joint, cosine schedule	0.660 ± 0.020	0.825 ± 0.033	0.359 ± 0.013

Two matched pairs, not a single-factor design. The four core arms do not all differ in one factor. Joint-linear and frozen-random use a linear reduction stage with normalization; frozen-pretrained and fine-tuned-pretrained use a pretrained multilayer spectral encoder without reduction-stage normalization. The clean freezing comparisons are therefore the two matched pairs, joint-linear against frozen-random and fine-tuned against frozen-pretrained. Comparing across the pairs — frozen-pretrained against frozen-random, for instance — changes architecture, initialization and training history, and validation selection, not only informativeness.

Verdicts. All comparisons in Table 25 are paired by seed with 90% t-intervals ($t = 2.920$, $n = 3$), conditional on one validation split, and are descriptive seed summaries rather than a population or multi-factor causal analysis.

Table 25: Matched protocol: paired differences with 90% paired-t intervals ($n = 3$). The last two rows compare the two protocols and are reported as descriptive seed summaries; saving a checkpoint changes what is available to evaluate, not the final-window trajectory.

Contrast	Final-5	Best-val
Fine-tuned − frozen-pretrained	$+0.021$ $[-0.022, +0.063]$	-0.004 $[-0.048, +0.039]$
Joint-linear − frozen-random	$+0.011$ $[-0.142, +0.163]$	$+0.023$ $[-0.066, +0.112]$
Frozen-pretrained − frozen-random	—	$+0.047$ $[-0.042, +0.136]$
Spectral lr $\times 10 - \times 1$	-0.133 $[-0.419, +0.152]$	$+0.048$ $[-0.016, +0.112]$
Spectral lr $\times 0.1 - \times 1$	-0.068 $[-0.173, +0.037]$	$+0.004$ $[-0.039, +0.046]$
Joint/frozen gap: matched − original	$+0.120$ $[-0.230, +0.470]$	-0.010 $[-0.105, +0.085]$

The pre-registered prediction that fine-tuning a pretrained spectral encoder falls below leaving it frozen is not supported: neither difference is resolvable at $n = 3$, and the fine-tuned arm moved $\boldsymbol{\theta}$ by only 2% of its initial norm. The logged applied update norms verify nonzero applied encoder updates under AdamW in that arm; this is a statement about the applied updates, not an attribution of the loss decrease. The pre-registered harmful-drift ordering $\times 0.1 \geq \times 1 \geq \times 10$ is not observed: on best-validation the $\times 10$ arm is above $\times 1$ in $3/3$ seeds while its final-window endpoint is the lowest and most variable of any arm at 0.561. Three multiplier means do not establish a systematic relation between displacement and peak score, and the intervals are wide. The cosine schedule improved neither mean endpoint (0.825 best-val and 0.660 final-5 against 0.837 and 0.694 for the constant learning rate); one schedule that did not help does not rule out an excessive late learning rate.

Disclosures. Width 48 only; $n = 3$; a single validation fold of 28 fold-0 cores; no test set. The best-validation column is exploratory and the final-window column is the pre-specified metric. The pretrained encoders were selected by pixel validation macro-F1 on the *same* fold-0 validation cores the runner later scores, which is a selection advantage for both pretrained arms, and were pretrained with weight decay 0.01 on $\boldsymbol{\theta}$. Pixel pretraining used the unaugmented centre-crop protocol while subsequent training used random crop plus augmentation; this affects comparisons across the two pairs, not the frozen-versus-fine-tuned pair that starts from the same checkpoint. Augmentation is on in all arms. Finally, the summary artifact `results/e3c_analysis_runs.csv` does not expose per-run configuration hashes or data-order pairing, so before the pairs are described as completely matched a compact provenance manifest should accompany it, recording architecture, initial checkpoint hash, parameter-group membership, clipping and weight-decay settings, normalization mode and state, augmentation and data-order seed, scheduler and selection rule; identical seed labels alone do not establish bit-identical mini-batch order across different model constructions.

E.1.7 What is not certified on this pipeline

The spatial module's input-Jacobian operator norm grows over training, from 3.2 to 157 on average for joint-linear (up to 414) and from 1.7 to 1.2×10^4 for joint-MLP, which rejects an initialization-scale stability approximation for the containment constant of the displacement theorem, though not the existence of a finite-horizon constant over the training window. The fitted per-step residual-envelope rate for joint-linear grows from $\hat{\mu} = 6.0 \times 10^{-4}$ ($M = 48$) to 1.9×10^{-3} ($M = 192$), a factor 3.2 for a $4\times$ width increase; the ratio is invariant to the step-clock-to-flow-time conversion, the absolute values are not, and two widths do not establish a trend.

F Verification record

The statements of this paper rest on the following checks, each a script in the accompanying repository with fixed seeds and a recorded outcome; the record is scoped to these statements and does not certify anything else. (i) Theorem 1: the closed forms, bounds and the normalized-readout control were verified against a full $(2 + M)$-parameter integration of the flow over 525 non-trivial initialization–width–margin cases and a $4{,}000$-draw Gaussian initialization sweep (`code/experiments/toy_serial_ce.py`, `toy_serial_ce_isotropic.py`; Appendix B); the algebraic identities of the proofs were checked symbolically and numerically (`code/audits/astra_math_02_checks.py`). (ii) Experiment 2: the matrix-product implementation of the convolutional head agrees with `nn.Conv2d` with circular padding to 10^{-15} in double precision, in outputs and parameter gradients, at widths 2, 37 and 512 (`exp2_intervention.py -selftest`); the calibration is bitwise reproducible before and after the paired-condition change to the generator; the pre-registered constants, thresholds, predictions and every amendment are dated in `paper/REFOCUS_PLAN_2026-09-10.md`. (iii) Experiment 3: preprocessing, principal directions and the readiness scan are computed on training patients only and evaluated on a core-disjoint half; no test-patient quantity was used for any choice. (iv) The production-model measurements of Appendix E carry their own verification record from the earlier draft (`code/audits/README.md`); only the statements quoted in Section 6 are relied upon here.